

High Throughput Bioprinting Using Decellularized Adipose Tissue-Based Hydrogels for 3D Breast Cancer Modeling

Priyanshu Shukla, Ashis Kumar Bera, Sriya Yeleswarapu, and Falguni Pati*

3D bioprinting allows rapid automated fabrication and can be applied for high throughput generation of biomimetic constructs for in vitro drug screening. Decellularized extracellular matrix (dECM) hydrogel is a popular biomaterial choice for tissue engineering and studying carcinogenesis as a tumor microenvironmental mimetic. This study proposes a method for high throughput bioprinting with decellularized adipose tissue (DAT) based hydrogels for 3D breast cancer modeling. A comparative analysis of decellularization protocol using detergent-based and detergent-free decellularization methods for caprine-origin adipose tissue is performed, and the efficacy of dECM hydrogel for 3D cancer modeling is assessed. Histological, biochemical, morphological, and biological characterization and analysis showcase the cytocompatibility of DAT hydrogel. The rheological property of DAT hydrogel and printing process optimization is assessed to select a bioprinting window to attain 3D breast cancer models. The bioprinted tissues are characterized for cellular viability and tumor cell-matrix interactions. Additionally, an approach for breast cancer modeling is shown by performing rapid high throughput bioprinting in a 96-well plate format, and in vitro drug screening using 5-fluorouracil is performed on 3D bioprinted microtumors. The results of this study suggest that high throughput bioprinting of cancer models can potentially have downstream clinical applications like multi-drug screening platforms and personalized disease models.

must be specific to the process or target tissue being mimicked to provide a suitable microenvironment to target cells.^[3] Extracellular matrix (ECM) provides microarchitecture and structural integrity to the tissue and can serve as a natural biomaterial source.^[4] Tissue decellularization allows for deriving tissue-specific ECM, an attainable goal for its use as a biomaterial in developing porous 3D scaffolds and decellularized ECM (dECM) hydrogels for tissue engineering applications.^[5,6] Decellularized tissue offers a variety of advantages over other natural biomaterials—essential native ECM components and growth factors, providing native tissue-specific microenvironment and architecture, retention of mechanical properties as observed in vivo, the capability of host integration and low immunogenicity given that cellular and genetic material is almost entirely removed during decellularization protocol.^[7,8]

Adipose tissue forms a supportive cushion around surrounding tissues and organs and acts as an energy storage material along with other metabolic functions.^[9] It carries a complex network of ECM proteins, which includes collagen type 1, 3, 4, 5, and 6; elastin, laminin, fibronectin as well as

proteoglycans, and other bioactive soluble factors such as VEGF and bFGF.^[10,11] Physiologically, subcutaneous adipose tissue carries differentiated adipocytes or fat cells that comprise 90% of the total tissue volume and 15% of the total cells within the tissue.^[12] The adipocytes are usually found in proximity to the vascular network since adipogenesis and angiogenesis occur in synergy; additionally, adipocytes are supported by a basement membrane and stromal cells.^[13,14] In its clinical applications, human adipose tissue is usually discarded as medical waste during surgeries like lipectomy or breast reduction, although it can be processed and used as a resource for adipose tissue engineering.^[15,16] Decellularization of adipose tissue has been previously reported using a combination of physical, chemical, and biological processes such as freeze-thaw cycles, trypsin digestion, detergent-based treatment of tissue using SDS with Triton X-100 as well as non-detergent treatments, followed by polar solvent treatment, and DNase/RNase treatment.^[17] There are a range of protocols available, but each has its drawbacks, such as time-consuming protocol due to two or more decellularization treatments carried out together, batch-to-batch variations due to different biological

1. Introduction

3D bioprinting technology has proven pivotal in addressing research questions in tissue engineering and regenerative medicine. By automated deposition of cell-laden hydrogels at predefined positions, 3D bioprinting has also paved the way for modeling cancer progression and applications for drug screening.^[1] Biofabrication of 3D tumor constructs has helped make relevant biomimetic platforms for rapid drug screening and reaching a step closer to translational research.^[2] A critical aspect of 3D bioprinting is the selection of biomaterial, which

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specimens, and failure of preservation of native adipose tissue structure due to loss of ECM components such as laminin or retention of lipid due to harsh surfactant treatments, and have been summarized well in previous reports.^[5,17] Therefore, there is always a need to develop an efficient decellularization method.

Decellularized adipose tissue (DAT) has been extensively used in the medical field, specifically reconstructive surgeries, to correct structural issues, commonly known as cosmetic surgeries.^[16,18] A common strategy is to fabricate scaffolds made of patient-derived DAT and then reseed the scaffold with patient-derived adipose-derived stem cells, and this construct is used as a graft to be implanted in patients for surgeries.^[19,20] The major limitation of using human-origin DAT is that it requires laborious manual handling. This prevents a widespread application of DAT due to questionable safe tissue preparation and the unavailability of a sufficient volume of allogenic tissues for clinical purposes.^[21] DAT can be used for in vitro tissue engineering and in vivo adipose tissue regeneration.^[16,18,22–26] There are numerous reports of the successful use of DAT in the field of tissue engineering and regenerative medicine with the development of DAT-based scaffolds and hydrogels. DAT has been used for bone regeneration,^[24,27] breast reconstruction,^[28] and other such tissue engineering purposes^[29–32]; however, there are a limited number of studies that have been able to repurpose DAT as bioink material for extrusion-based 3D bioprinting.^[33,34]

DAT has also found its application in the field of cancer biology and DAT-based 3D scaffolds have been used for various cancer studies. Castello-Cros et al. used a fibroblast-produced ECM scaffold to culture and study breast cancer cell lines.^[35] Similarly, Hoshiba & Tanaka used breast cancer cell line-produced ECM scaffolds to elucidate breast cancer cell behavior on these tumorigenic matrices.^[36] Dunne et al. used human adipose tissue-derived ECM 3D scaffolds to encapsulate breast cancer cells MCF-7 and BT474 breast cancer cells to study tumor progression and elucidate response to chemotherapeutic drugs. The study showed that in vitro cellular proliferation within DAT scaffolds significantly matched the proliferation profile of cells within in vivo xenografts compared to results from 2D and Matrigel cultures.^[37] Jin et al. performed a long-term culture of MCF-7 breast cancer cells on decellularized breast cancer biopsies.^[38] Liu et al. did an extensive biological characterization of MCF-7 culture on decellularized breast cancer biopsies.^[39] Ruud et al. characterized breast cancer cell behavior on dECM and matrigel scaffolds.^[40]

However, only a few studies have been performed using DAT-based hydrogels to fabricate breast cancer models using extrusion-based 3D bioprinting technology. A recent study where DAT hydrogel was used to bioprint a 3D breast cancer model was proposed by Chen et al. The bioink comprised of MCF7-laden decellularized porcine fat mixed with GelMA was subjected to extrusion-based 3D bioprinting of multilayered lattice constructs.^[41] However, the printed constructs were monoculture and unsuitable for downstream clinical applications that require a biomimetic, high-throughput testing system. A 3D bioprinted breast cancer model was put forth by Blanco-Fernandez et al., wherein the group formulated decellularized porcine breast tissue (TDM) mixed with multiple other biomaterials including GelMA and alginate to increase the printability of the bioink along with the introduction of type 1 collagen. MCF7 and

hAMSCs-laden bioink was printed in a multilayered scaffold with a cancerous core and an outer stromal layer.^[42] However, because of the additional biomaterials, the biological effect of the dECM bioink on cells alone could not be independently studied.

With an increasing focus on 3D culture systems and their application as screening platforms, an essential feature while designing novel 3D models is their ability to conduct high throughput assays. The rise of 3D bioprinting technology has led to the development of physiologically relevant models to study organ biology, disease pathophysiology, and its progression, as well as to address different tissue engineering research questions. Notably, high throughput platforms can be biofabricated to address the needs of pharmaceutical industries. An important observation based on previously mentioned studies is that most studies developed a bioprinted 3D lattice or simple lattice structure. Some of these models maintained a balance between complexity and architecture but compromised on developing these models as high throughput testing platforms by using traditional printing approaches. With the help of the 3D bioprinting technique, biofabrication of 3D scaffolds can be done rapidly in a high throughput manner instead of manual pipetting, increasing sample size and the number of experimental assays and allowing for screening of multiple drugs.

This study presents a high throughput method for bioprinting high-density microtumors utilizing DAT-based bioink for breast cancer modeling. A method for detergent-free decellularization of caprine-origin adipose tissue for bioink formulation was developed, and we fabricated lattice-like bioprinted cancer models and characterized them for cancer cell viability and functionality. Unlike previous studies, the proposed caprine-origin dECM-based bioink did not require another viscosity modifier for bioprinting. Additionally, we modified the printing protocol to develop a rapidly fabricated, high-throughput 3D bioprinted platform for drug screening applications using a reduced amount of biomaterial within 7 min for printing high cell density constructs on a 96-well plate. We further validated the 3D bioprinted microtumors via anti-cancer drug cytotoxicity assessment. Overall, our study provides the application of DAT hydrogel for the development of a high throughput cancer model and mainly has an important implication for the fabrication of personalized cancer models using patient cells for rapid drug screening in clinical settings.

2. Results

2.1. Decellularization of Caprine Adipose Tissue

The process of decellularization involves the removal of cellular content from tissue ECM using different chemicals and detergents. The end product is a biomatrix that provides a microenvironment devoid of cells with a 3D organization similar to native living tissues. Here, we have used three different methods to decellularize caprine omental fat. **Figure 1A** shows the different decellularization steps to obtain the final lyophilized DAT tissue. We used three protocols to decellularize the white adipose tissue: the SDS-Triton detergent-based method and two salt-based methods—hypo-hyper (HH) and distilled water-hyper (WH). We investigated the resultant tissue and found that the detergent-based method led to increased lipid removal; however, a loss of tissue matrix was seen as compared to salt-based methods. We

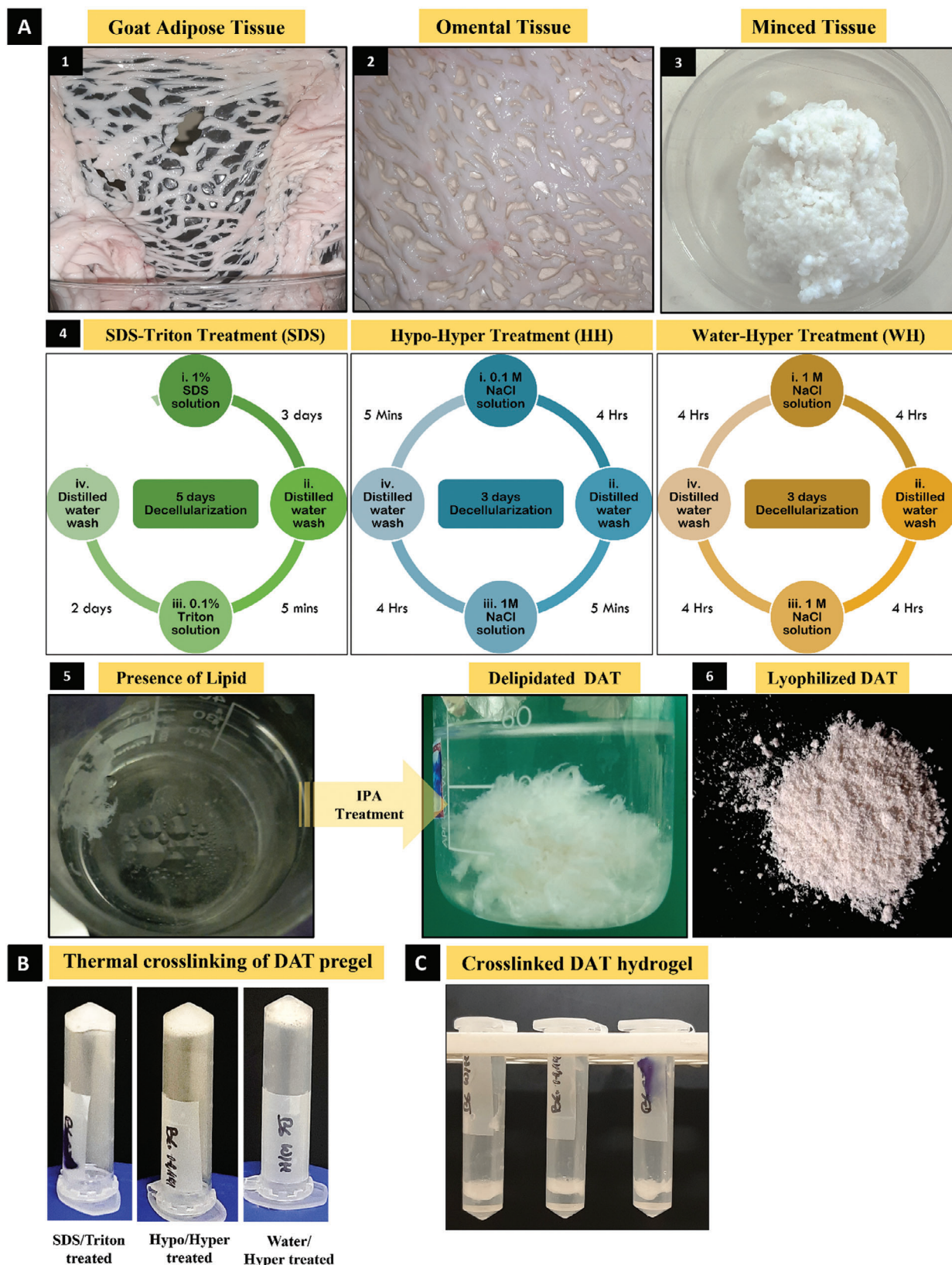


Figure 1. Preparation and decellularization of decellularized adipose tissue: A1,2) caprine adipose tissue and tissue isolated for decellularization, A3) minced adipose tissue, A4) decellularization processes using one cycle of detergent and salt-based methods, A5) delipidation, A6) after lyophilization. B) pH-adjusted digested DAT hydrogel after 45 min of thermal crosslinking at 37 °C. C) DAT hydrogel stability in PBS at 37 °C.

Table 1. A summary of steps for decellularization used in the study.

Day	Processing Stages
1	Wash omental adipose tissue, chop it into smaller pieces, and incubate in detergent and non-detergent decellularizing solutions.
2, 3, 4	Multiple decellularization washes followed by a distilled water wash (30 min).
5, 6	Multiple washes with 100% IPA for delipidation (minimum of 48 h till complete lipid removal) followed by a distilled water wash.
7	Tissue incubation was done overnight in 3% H ₂ O ₂ solution (12 h).
8, 9	Multiple distilled water washes were given, and samples were incubated at −20 °C until lyophilization.
10, 11	Lyophilization of samples for 48 h until complete dry tissue is obtained.

found a significant amount of white fat attached to tissue matrix post-decellularization washes; therefore, it was followed by isopropanol treatment to remove associated lipids for a minimum of 48 h until no presence of lipid was observed (Figure 1A(1–5)). We observed a higher fat content in a few batches, for which we consequently increased the duration of IPA treatment up to another 48 h until complete delipidation was observed.

Post lyophilization, the ECM was subjected to acetic acid for pepsin-based enzymatic digestion to solubilize collagen fibrils to a final concentration of 3% (w/v), and then the pH was adjusted at a low temperature to a neutral pH of 7.4. All the pH-adjusted DAT hydrogels could be stored at 4 °C for up to 2 weeks till further experiments optimally. To attain structurally stable constructs, viscous pH-adjusted DAT hydrogels were thermally crosslinked at 37 °C to form 3D DAT hydrogel scaffolds (Figure 1B). Crosslinked DAT scaffolds stayed stable in PBS solution, indicating the stability of crosslinked constructs (Figure 1C).

A summary of the steps used for the decellularization protocols in this study is presented in Table 1.

2.2. Physical and Biochemical Characterization

Post-decellularization, the physical characteristics of lyophilized tissues obtained from three methods were analyzed. To understand the fiber morphology of decellularized tissues, we applied SEM imaging analysis to characterize them. The ECM fibrils within all the decellularized samples were well preserved and highly condensed, as seen in Figure 2A, with finer fibers in WH DAT compared to SDS DAT or HH DAT.

Biochemical Characterization of lyophilized DAT was performed to quantify the retention of ECM components like collagen and sulfated GAGs. It was observed that 59.246% ± 4.133%, 62.385% ± 2.884%, and 33.005% ± 1.323% of hydroxyproline content was retained in SDS DAT, HH DAT, and WH DAT, respectively, compared to native adipose tissue. Meanwhile, 82.746% ± 5.472%, 80.606% ± 1.647%, and 72.462% ± 0.462% sGAGs were retained in SDS DAT, HH DAT, and WH DAT, respectively, compared to native adipose tissue, as seen in Figure 2B. Decellularization efficiency was assessed by quantifying the DNA content left over in lyophilized tissue compared to native tissue. We initially noticed that the native adipose tissue was not wholly solubilizing during papain digestion, and a

minute DNA content was observed in the assay. Therefore, we introduced additional IPA wash for 48 h to native tissue to remove lipids before taking that sample for papain digestion. Compared to native adipose tissue, we saw a significant DNA content reduction from 902.546 ± 1.123 ng mg^{−1} to 24.6 ± 17.383 ng mg^{−1} in SDS DAT lyophilized tissue, 49.52 ± 26.538 ng mg^{−1} in HH DAT lyophilized tissue, 43.64 ± 25.253 ng mg^{−1} in WH DAT lyophilized tissue as seen in Figure 2C. All the values were below the standardized benchmark of DNA content of 50 ng mg^{−1} of lyophilized decellularized tissue, as reported previously,^[17,20,34,43,44] which means the tissue is considered as efficiently decellularized and these biomaterials will not elicit any immune response.

2.3. Histological Characterization of DAT

DAT obtained using the HH protocol of decellularization (HH DAT) (Figure 2E) was subjected to further characterization to assess the efficacy as compared to native adipose tissue (Figure 2D). Nuclear staining of tissue sections using Hoechst 333258 showcased that most of the nuclei had been removed from HH decellularized tissue (lower panel) compared to native adipose tissue (upper panel). Oil Red-O staining revealed the absence of lipids in HH DAT (lower panel) compared to large deposits of lipids in native adipose tissue (upper panel).

2.4. Cytocompatibility of DAT Hydrogel-Based Scaffolds

We next investigated the cytocompatibility of DAT hydrogels and observed cell adhesion to the matrix and survival over a culture period of 7 days. We encapsulated MDA-MB-231 breast cancer cells within 3% HH and WH DAT hydrogel as test groups and rat tail collagen hydrogels as control. A live/dead assay was performed on cell-laden 3D hydrogel scaffolds to assess their cytocompatibility. Viable, live cells were stained with calcein-AM as visualized in green and the dead cells were stained with PI as visualized in red, as seen in fluorescence images in Figure 3A. Cells were homogeneously distributed through the hydrogel groups without significant clustering at day 1. The 3D collagen group was considered a control, and there was no significant difference in cellular viability for our DAT-based hydrogel groups compared to collagen hydrogel. All the DAT hydrogels at the day 7 time point showcased high cell viability above 95%, and results were quantified in Figure 3B. While collagen hydrogel showed 92.8% ± 1.411% cell viability on day 1 and 96.94% ± 1.558% on day 7, 3% HH DAT hydrogel showcased viability of 97.83% ± 1.624% at day 1 and 99.27% ± 0.907% day 7; even the derivative WH DAT hydrogel had a cell viability of 97.66% ± 1.954% at day 1 and 99.41% ± 0.791% at day 7.

Figure 3C showcases the results of the MTT assay where an increased metabolic activity was seen over the culture period for the cell-laden 3% HH DAT hydrogel group in a time-dependent manner, with a significant increase in metabolic activity at day 7 as compared to day 1, showcasing cell supporting properties.

As a preliminary test, we also wanted to assess the response of different types of cells to our DAT-based hydrogel. We cultured

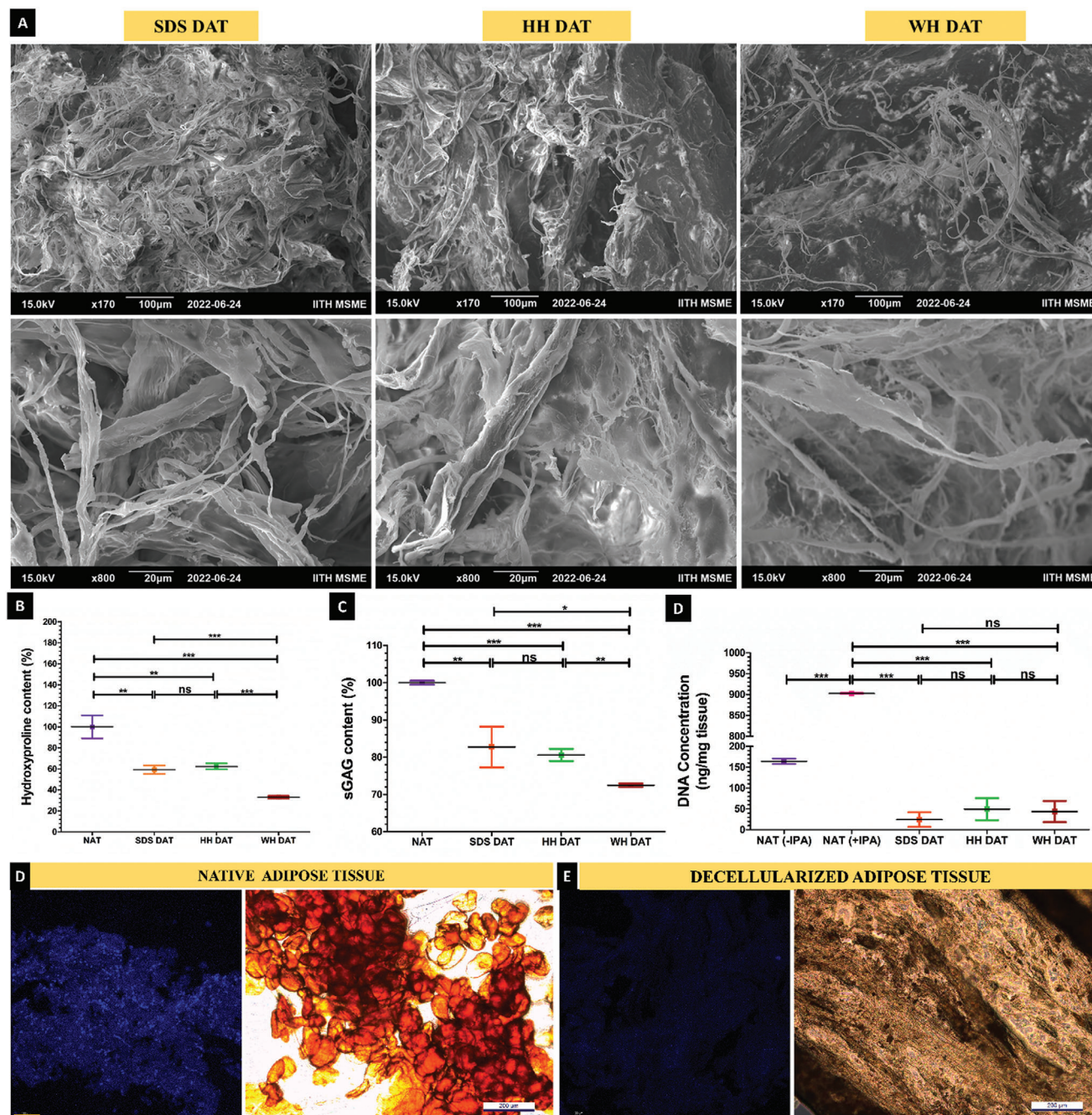


Figure 2. Physical and biochemical characterization of decellularized tissues: A) SEM analysis to showcase surface morphology of decellularized tissues. B) Quantification of the content of Hydroxyproline and sGAG content in native adipose tissue and decellularized tissues. C) DNA quantification of native adipose tissue and decellularized tissues. D, E) Hoechst 33258 and ORO staining of D) native adipose tissue and E) HH DAT. Scale bar size: A) Upper panel 100 μ m, lower panel 20 μ m, D, E) Hoechst 33258 stained images 100 μ m, ORO-stained images 200 μ m. $n = 3$, $*p \leq 0.05$, $**p \leq 0.01$, and $***p \leq 0.001$.

4T1 breast cancer cells, L929 fibroblast cells, and primary limbal stem cells with 3% HH DAT hydrogel. We observed that once cells had entirely populated the scaffolds, the cells started to show stretched, spindle-like phenotypes at the edges of scaffolds and, to a certain extent, within the scaffold, although the time taken for cells to adjust to the matrix varied for all the cells. Figure 3D–F shows microscopic images from day 7 for 4T1 cells,

day 10 for L929 cells, and day 4 for limbal stem cells. As seen in Figure 3G,H showcases the results of the MTT assay where an increased metabolic activity was seen over the culture period for cell-laden 3% HH DAT hydrogel with a significant increase in metabolic activity from day 7 as compared to day 1, showcasing cell-supporting properties. This indicates that the DAT hydrogels can support and develop heterogeneous in vitro models.

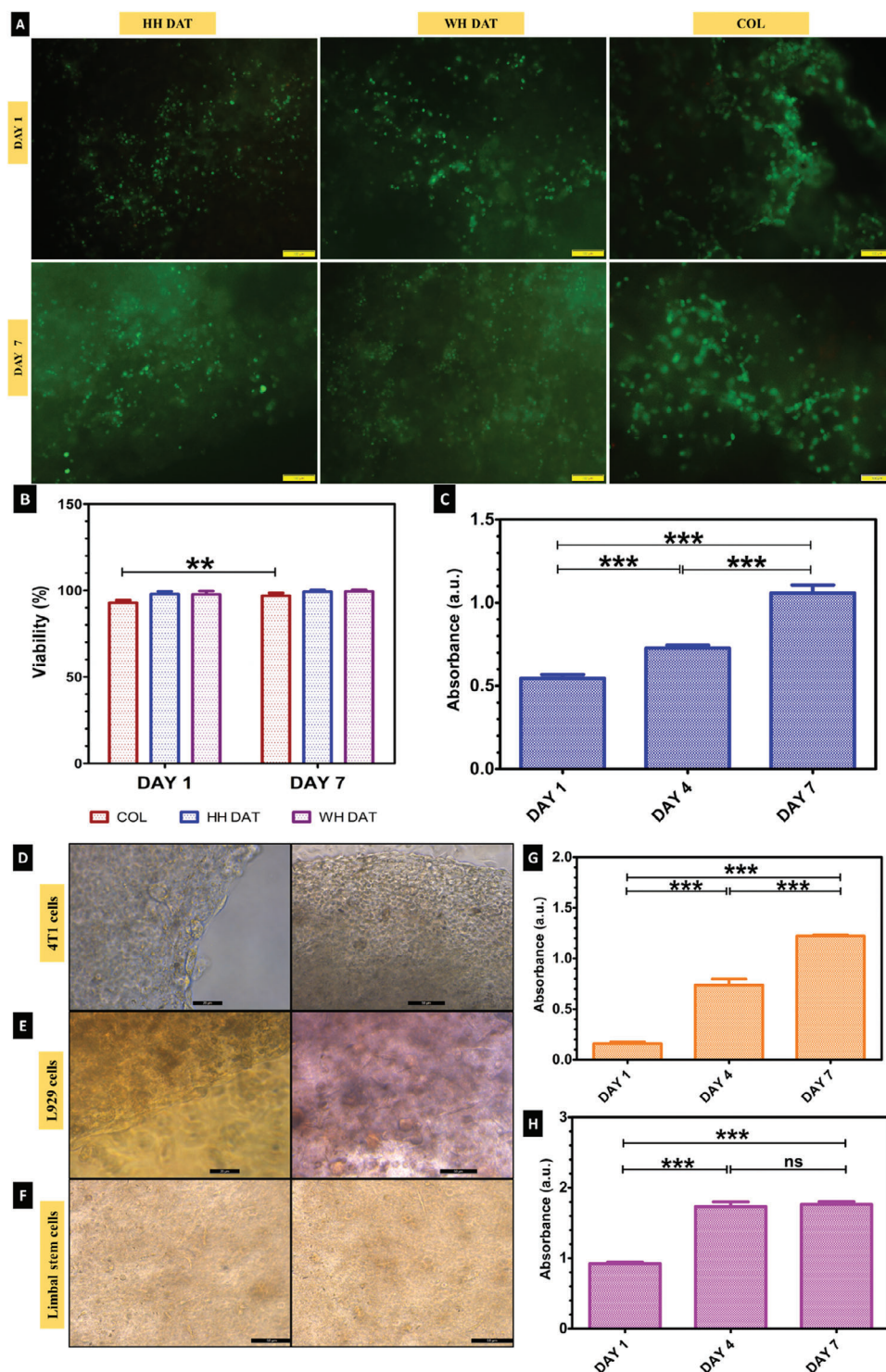


Figure 3. Biological Characterization of DAT Hydrogels: A) Live/dead assay to depict MDA-MB-231 breast cancer cell viability in 3% HH DAT, 3% WH DAT, and collagen (1 mg mL^{-1}) hydrogels at days 1 and 7 (viable cells are in green and dead cells are in red). B) Quantification of live/dead assay results. C) Comparison of metabolic activity of MDA-MB-231 cells at different time points in HH DAT hydrogel. D–F) Comparison of cell morphology in encapsulated cells within 3% HH DAT scaffold. Phase contrast images of D) 4T1 breast cancer cells, E) L929 fibroblast cells, F) Limbal stem cells. Comparison of metabolic activity of G) 4T1 cells and H) L929 cells at different time points in HH DAT hydrogel. Scale bar size: A) $100 \mu\text{m}$ and D–F) $200 \mu\text{m}$. $n = 3$, $*p \leq 0.05$, $**p \leq 0.01$, and $***p \leq 0.001$.

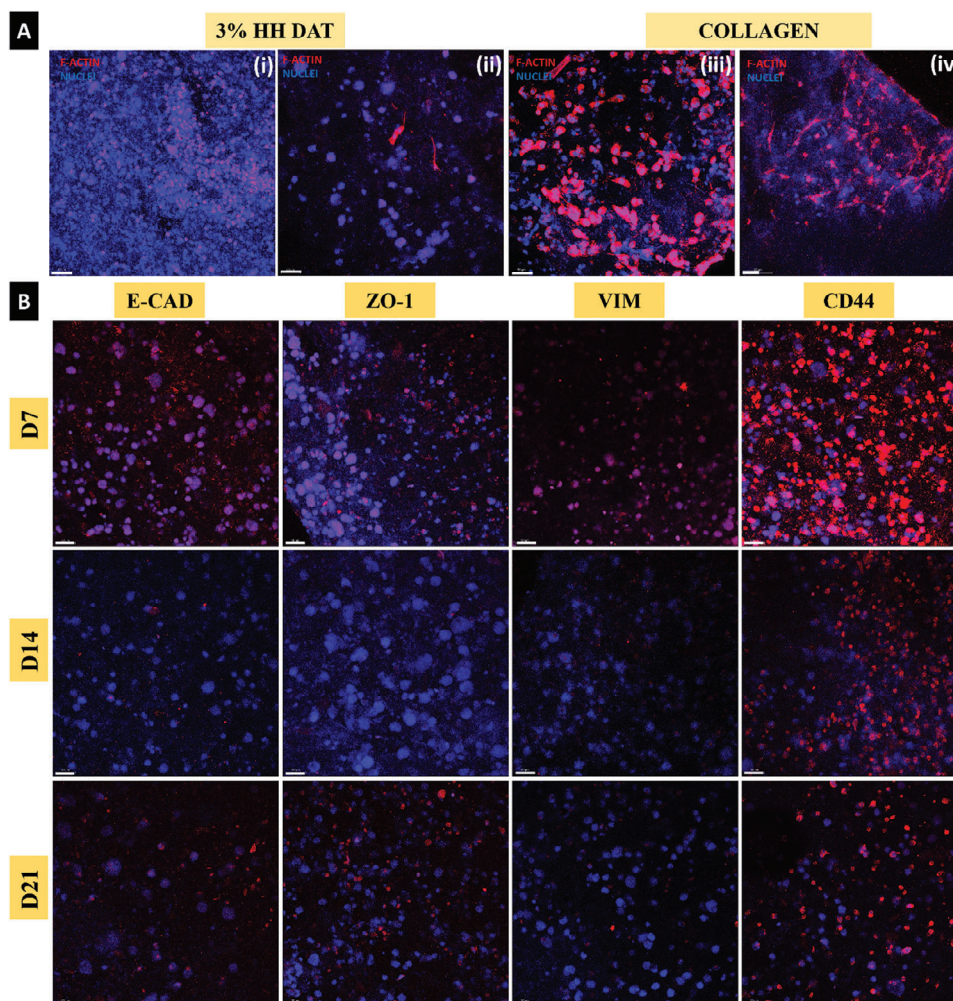


Figure 4. Assessment of functional marker expression in cell-laden DAT scaffold. A) Immunofluorescence images showing F-actin expression in encapsulated MDA-MB-231 cells on i,iii) day 1 and ii,iv) day 7 of culture within HH DAT scaffold. (Functional marker is in red, and nuclear content is stained in blue). B) Immunofluorescence images showing the expression of E-Cad, Zo-1, Vim, and CD44 in encapsulated MDA-MB-231 cells on days 7, 14, and 21 of culture within HH DAT scaffold. (All functional markers are in red; nuclear content is stained in blue). Scale bar size: A,B) 50 μ m.

2.5. Cellular Functionality of DAT Hydrogel

2.5.1. Assessment of Cell Morphology

Next, we used confocal microscopy to assess the cellular morphology of MDA-MB-231 breast cancer cells encapsulated within HH DAT-based 3D scaffolds. We could observe the cell morphology of encapsulated cells on days 1 and 7 of the culture, as shown in **Figure 4A**. The thickness of hydrogels was confirmed to be 400 ± 100 μ m (representative image included as Figure S1, Supporting Information) via Z-stacking using confocal imaging, and cells were viable and proliferating during the culture period. Separate figures for cells cultured in scaffolds at different time points are shown in Figure S2, Supporting Information. The 3% HH DAT hydrogel groups showed the majority of spherical MDA-MB-231 cells forming aggregates by day 7 as compared to cells encapsulated within collagen hydrogel, where a mixture of spherical and spindle-shaped, epithelial-like cells could be seen. The cells

proliferated within the hydrogels over the culture period from days 1 to 7 in all hydrogels.

2.5.2. Assessment of Functional Markers

Previous experiments showed that HH protocol-based decellularized adipose-based matrix is a viable biomaterial showing good cytocompatibility. Next, we wanted to investigate the behavior of MDA-MB-231 breast cancer cells encapsulated within the 3D decellularized adipose matrix-based scaffold and assess their functionality within a 3D microenvironment. With that aim, we checked for the expression of multiple cell surface markers like E-Cadherin, ZO-1, Vimentin, and CD-44. Large-sized cancer cell clusters or microspheres were observed toward the edges of scaffolds compared to within the scaffolds. It was observed that cancer cell clusters and the population of cells as single and isolated entities coexisted throughout the culture period, as seen in

Figure 4B. Separate Figures for each marker at different time points of culture are shown in Figure S3, Supporting Information. First, regarding E-Cadherin, a cell surface receptor known for cell-to-cell adhesion, and ZO-1, a tight gap junction protein, it was observed that their expression levels were gradually lowered for cells present within the cell clusters across the culture period. Expression for another cell surface receptor, Vimentin, a mesenchymal marker, was seen in various cells through all time points; however, its expression was comparatively less as compared to E-Cadherin and ZO-1 with increasing culture period, indicating a more differentiated state of the cancer cells. Similarly, a CD44 expression profiling was done. It was observed that CD44 was expressed in most cells, and its expression could be seen around individual cells and microspheres through all time points.

2.5.3. Assessment of Hypoxia

We wanted to explore the reason for the presence of cell clusters developing over 21 days of culture within the 3D scaffolds as seen in previous experiments and hence assessed for expression of hypoxia marker HIF1 α within encapsulated MDA-MB-231 cells. Using confocal microscopy, we could rule out hypoxia as a reason for cell clustering and ensure sufficient oxygen and nutrient diffusion within the 3D DAT hydrogel-based scaffolds. As seen in Figure S4, Supporting Information, the encapsulated MDA-MB-231 cells within 3% HH DAT scaffolds did not experience a hypoxic environment even after 21 days of cell culture, which was indicative of HH DAT hydrogel-based scaffold being stable and thoroughly permeable to nutrients and oxygen showcasing DAT hydrogel as suitable long-term culture biomaterial. The different-sized cell clusters forming within and at the edges of the scaffold also developed independently without any hypoxia-mediated mechanism.

2.6. Rheological Profile of DAT Hydrogel

We selected the HH DAT tissue for further experimentations based on previous experiments. We aimed to assess the flowability of HH DAT hydrogel as the hydrogel should maintain enough viscosity to be extruded out and retain its shape after forming a 3D scaffold. Different rheological properties were studied to characterize the flow behavior property of HH DAT hydrogel. Shear rate sweep for 3% w/v HH DAT hydrogel showcased decreasing viscosity with increasing shear rate, indicating the hydrogel's shear thinning behavior as seen in Figure 5A. An amplitude sweep was performed with pH-adjusted DAT hydrogel at a temperature of 20 °C where a strain sweep from 0.01 to 100% at a frequency of 1 Hz, and it was found that the hydrogel had a higher storage modulus than the loss modulus, as shown in Figure 5B. Linear viscoelastic region (LVER) for the hydrogel was observed up to 4% of the shear strain, and a crossover point was detected at a shear strain of 50.26%. Figure 5C showcases the gelation kinetics of HH DAT pre-gel. The pH-adjusted DAT hydrogel was first subjected to a temperature ramp from 4 to 37 °C with a temperature ramp of 0.5 °C min⁻¹, and the plot shows that the complex modulus remained almost constant at lower temperatures

and then steadily started to increase after 23 °C. The gel behavior at lower temperatures also mimics the cell encapsulation conditions when cells are encapsulated in the hydrogel for 3D scaffold experiments or bioink preparation over ice at 4 °C. The cell-laden hydrogel is filled in a syringe barrel, and the complete cartridge is loaded into the printer pre-set at 15 °C. During the bioprinting process, the bioink is exposed to a range of cold temperatures for a duration, and hence, to mimic these experimental conditions, this rheological experiment was performed to assess the effect of temperature as well as time (each temperature for 2 min) on the hydrogel strength. The gelation kinetic study revealed a steady complex modulus of uncrosslinked HH DAT hydrogel at 4 °C, followed by a steady increase at the end of the temperature ramp till 37 °C, after which the gel was held at 37 °C for 1 h to replicate the thermal crosslinking. This experiment was representative of a mechanically weaker uncrosslinked pH-adjusted hydrogel getting crosslinked to form a stable 3D scaffold. At the end of 1 h, similar to in vitro thermal crosslinking conditions at 37 °C, the complex modulus increased by 12-fold, indicating a crosslinked DAT hydrogel construct.

2.7. Optimization of Process Parameter for Printability with DAT Hydrogel

We further wanted to assess the printability of HH DAT hydrogel for 3D printing-based applications. For optimization of 3D printing of HH DAT hydrogels, we prepared a filtered acellular bioink using pH-adjusted 3% w/v HH DAT hydrogel. An essential highlight through rheological studies was that 3% HH DAT hydrogel showed shear thinning behavior and preluded the printable nature of adipose tissue-based decellularized ECM (dECM)-based hydrogels which provides an implication for the use of HH DAT hydrogel for 3D bioprinted cancer model.

The rheological properties of 3% HH DAT hydrogels showcased that gel was extrudable, could hold stress due to wide LVE region, and had the necessary shear thinning property to make this biomaterial an optimal bioink for extrusion-based 3D printing as shown in previous studies.^[3,45] A lattice geometry was 3D printed (3D printer, Regemat) with size 15 × 15 × 1 mm³ for two layers, with each layer of height equivalent to nozzle diameter—250 and 400 μ m. Printing, specifically lattice geometry, allows for the optimization of hydrogel extrusion under varying printing speeds and pressures using a printing nozzle, leading to a printed structure with high resolution. Two printing parameters were optimized for the DAT hydrogel—nozzle diameter and flow speed to ensure the fidelity or resolution of printed strands. For optimization of 3D printing of 3% HH DAT hydrogel, its extrusion through a 250 μ m tapered nozzle was checked at a room temperature of 25 °C (Figure 5D(i,ii)). We printed constructs using different flow speeds of 0.5 and 0.75 mm⁻¹ s at the optimal pressure. It was noted that printed strands showed more spreading at a lower flow speed of 0.5 mm⁻¹ s and better strand fidelity for the bottom layer at an increased flow speed of 0.75 mm⁻¹ s; however, the second layer would not maintain strand fidelity and, therefore, did not form a stable construct.

Next, we assessed the extrusion of the material through a 400 μ m tapered nozzle for optimization of 3D printing of 3% HH DAT hydrogel (Figure 5D(iii,iv)). Three different flow speeds

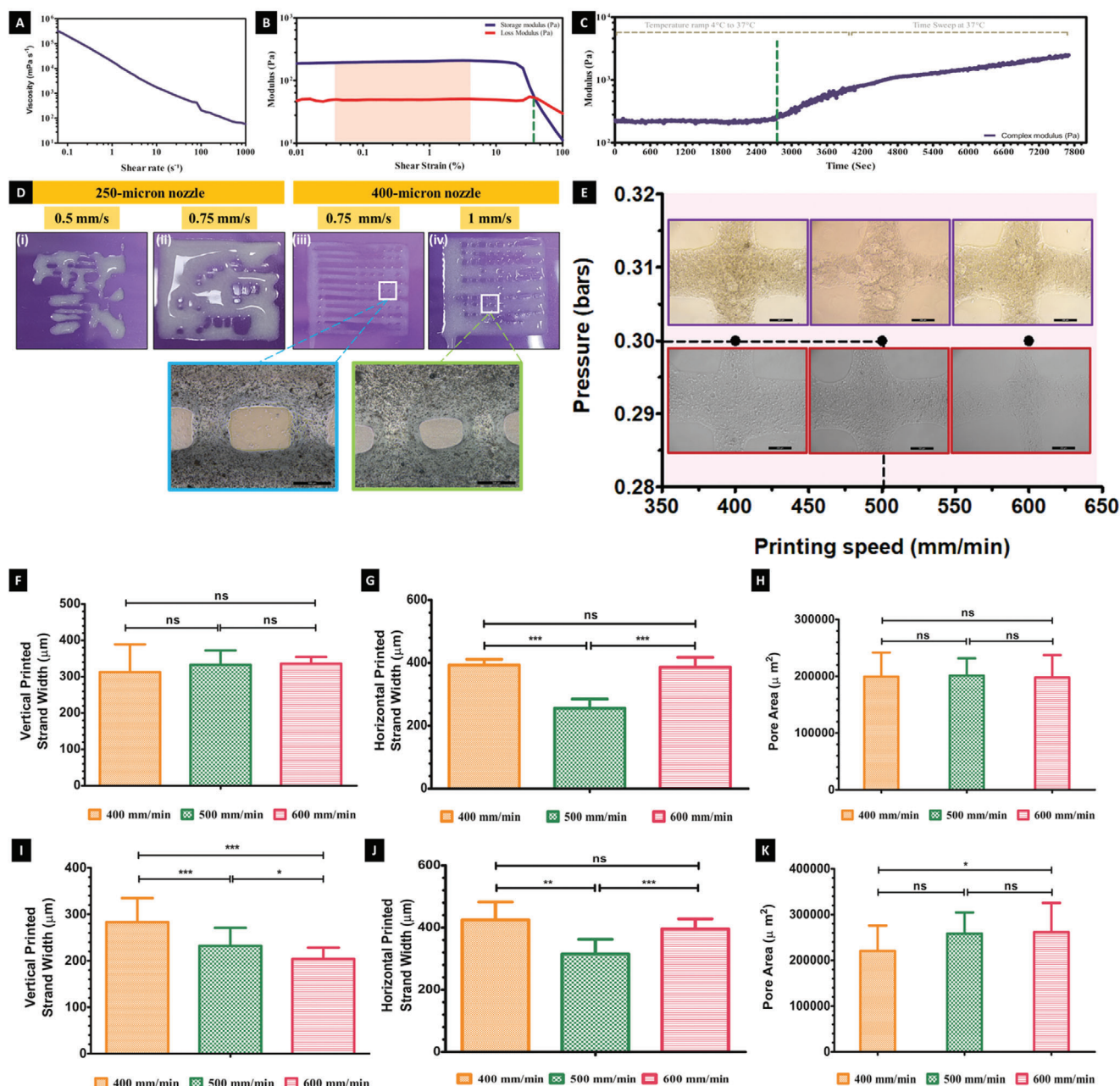


Figure 5. Optimization of extrudability of 3% HH DAT hydrogel. A) Rheological profile of HH DAT hydrogel with viscosity or shear rate sweep at 20 °C. B) The amplitude sweep indicates a linear viscoelasticity region (shaded region in pink). C) Gelation kinetics from 4 to 37 °C (started from 4 °C with a hold for 5 min at this temperature and with an increment of 0.5 °C min⁻¹ until reaching 37 °C and maintained at 37 °C for 60 min). D) Macroscopic images of 3D printed constructs with different nozzle diameters and printing speeds with inset microscopic images of 3D printed strands at 0.75 mm and 1.00 mm⁻¹ s. E) Characterization of TNBC bioprinted model (purple boxes) and 3D bioprinted TNBC stromal model (maroon boxes) developed with extrusion printing of bilayered lattice structures. Characterization of only MDA-MB-231 laden 3DBP constructs (upper panel; F–H) and MDA-MB-231+NIH-3T3 (1:1) laden 3DBP constructs (lower panel; I–K) F–I) Quantification and comparison of 3D printed vertical strand width within 3D bioprinted constructs. G–J) Quantification and comparison of 3D printed horizontal strand width within 3D bioprinted constructs. H–K) Quantification and comparison of pore area between horizontal and vertical printed filaments of 3D bioprinted constructs. Scale bar size: D) 500 μm for inset images and E) 200 μm . $n = 3$, * $p \leq 0.05$, ** $p \leq 0.01$, and *** $p \leq 0.001$.

were assessed—0.5, 0.75, and 1 mm⁻¹ s at room temperature of 25 °C. The hydrogel was not extruded uniformly at the flow speed of 0.5 mm⁻¹ s (figure not attached). It was observed that the strands extruded at a speed of 0.75 mm⁻¹ s showed the best-printed strand fidelity with minimal spreading, as seen in **Figure 6B**. The hydrogel was also extruded uniformly for the first layer at the flow speed of 1 mm⁻¹ s; however, some spreading was observed post-printing for the second layer. Therefore, the extrudability for 3% HH DAT hydrogel was observed and ascertained for its 3D printability for future experiments.

2.8. 3D Bioprinting of Heterogeneous Breast Cancer Model

2.8.1. Optimization of Bioprinting for TNBC Model

For 3D bioprinting, we first used an extrusion-based bioprinter (Dual nozzle Bioprinter, NBIL) to bioprint only MDA-MB-231 cells to form a 3D bioprinted TNBC model. A lattice geometry (**Figure 5E** upper panel) was bioprinted with a size of 15 × 15 × 1 mm³ for two layers, with each layer of height equivalent to a nozzle diameter of 250 μm. Three different printing speeds were used to 3D bioprint cancer models with DAT hydrogel against a constant pressure of 0.3 bars and at a temperature of 15 °C to optimize the fidelity or resolution of printed strands. We observed that among the printed constructs with three different printing speeds, the spreading of printed strands was lesser for strands printed at the speed of 500 mm min⁻¹ than compared to the other two speeds of 400 and 600 mm min⁻¹. Concerning morphological differences observed between all three printing speeds, no significant differences were seen within the 3D constructs (**Figure 5E**). **Figure 5F–H** showcased some degree of spreading occurring; however, it was under the acceptable range, and all the 3D printed constructs were stable and held their bilayer organization. However, it should be noted that the NBIL printer requires manual pressure adjustment at the start of every experiment and in case of nozzle clogging; therefore, that factor alone can influence the strand fidelity every time the experiment is replicated.

2.8.2. Optimization of Bioprinting for TNBC Stroma Model

Next, we used the extrusion-based 3D bioprinter to bioprint a bioink composed of both MDA-MB-231 cancer cells and NIH-3T3 fibroblast cells to attain a more relevant cancer model as breast tumors are known to have a stiff composition due to the presence of fibroblasts. Experimental conditions were kept the same for the comparative study. A lattice geometry (**Figure 5E** lower panel) was 3D bioprinted with size 15 × 15 × 1 mm³ for two layers, each layer of height equivalent to a nozzle diameter of 250 μm. Three different printing speeds were used to 3D bioprint cancer models with DAT hydrogel against a constant pressure of 0.3 bars to ensure the fidelity or resolution of printed strands. We observed that among the constructs fabricated at three different printing speeds, the printing resolution looked better at 600 mm min⁻¹ compared to the other speeds. Concerning morphological differences observed between all three printing speeds, no differences were seen in the 3D constructs (**Figure 5E**). **Figure 5I–K**

showcases some degree of spreading occurring at lower speeds of 400 and 500 mm min⁻¹ compared to 600 mm min⁻¹; however, it was under the acceptable range. All 3D printed constructs were stable during printing and held their bilayer organization after printing. Consequently, we 3D bioprinted using all three speeds, and the constructs attained were subjected to further biological characterization to assess if any differences arose due to printing speed.

2.8.3. Biological Characterization of 3D Bioprinted Breast Cancer Models

We next characterized the cellular viability of bioprinted constructs and observed cell adhesion to the matrix and survival over a culture period of 7 days for both the monoculture and coculture models. A live/dead assay was performed on the constructs printed using different speeds to assess cell viability within constructs printed using different speeds. Viable, live cells were stained with calcein-AM in green and dead cells were stained with PI in red, as shown in fluorescence images in **Figure 6A,B**. Cells were homogeneously distributed throughout the bioprinted constructs on day 2 for both constructs. A highly stretched network of fibroblasts along with breast cancer cells was seen in coculture constructs in the 3D bioprinted coculture model. In contrast, breast cancer cells noticeably started to come together in contact and form small clusters in the 3D bioprinted monoculture model by day 7 of culture compared to isolated entities on day 2. All the bioprinted constructs at the day 7 time point showcased high cell viability above 95%, and results were quantified in **Figure 6C,D**.

The extrusion-based 3D bioprinting of our fabricated cancer models was cell supportive, with average cell viability for monoculture constructs at 94.008% ± 2.793% on day 2 and 95.027% ± 4.802% on day 7 for all printing speeds and average cell viability for coculture constructs at 86.138% ± 4.91% on day 2 and 98.004% ± 0.529% at day 7 for all printing speeds. **Figure 6C,D** shows the quantified cell viability data from live/dead assay showcasing high cell viability post-printing process across a culture period of day 7. No significant difference in cell viability between tested printing speeds was observed for the 3D bioprinted breast cancer models—monoculture and coculture.

2.8.4. Assessment of Functional Markers in the 3DBP Cancer Model

Since the live/dead assay exhibited the presence of cell stretching within bioprinted constructs, we next assessed the expression of the cytoskeletal protein F-actin indicative of the cellular morphology of cells encapsulated within 3D bioprinted monoculture constructs using confocal microscopy. We could observe the cell morphology of encapsulated cells on day 7 of culture, as shown in **Figure 6E**. Over the culture period, both round clusters and spindle-shaped MDA-MB-231 cancer cells were present in all 3D bioprinted constructs as the breast cancer cells gradually adapted to the matrix and showed a stretched, spindle-shaped morphology for the duration of the culture. We also investigated the functionality of MDA-MB-231 breast cancer cells encapsulated within 3D bioprinted matrices to understand their response to the decellularized adipose matrix. With that aim, we checked for the

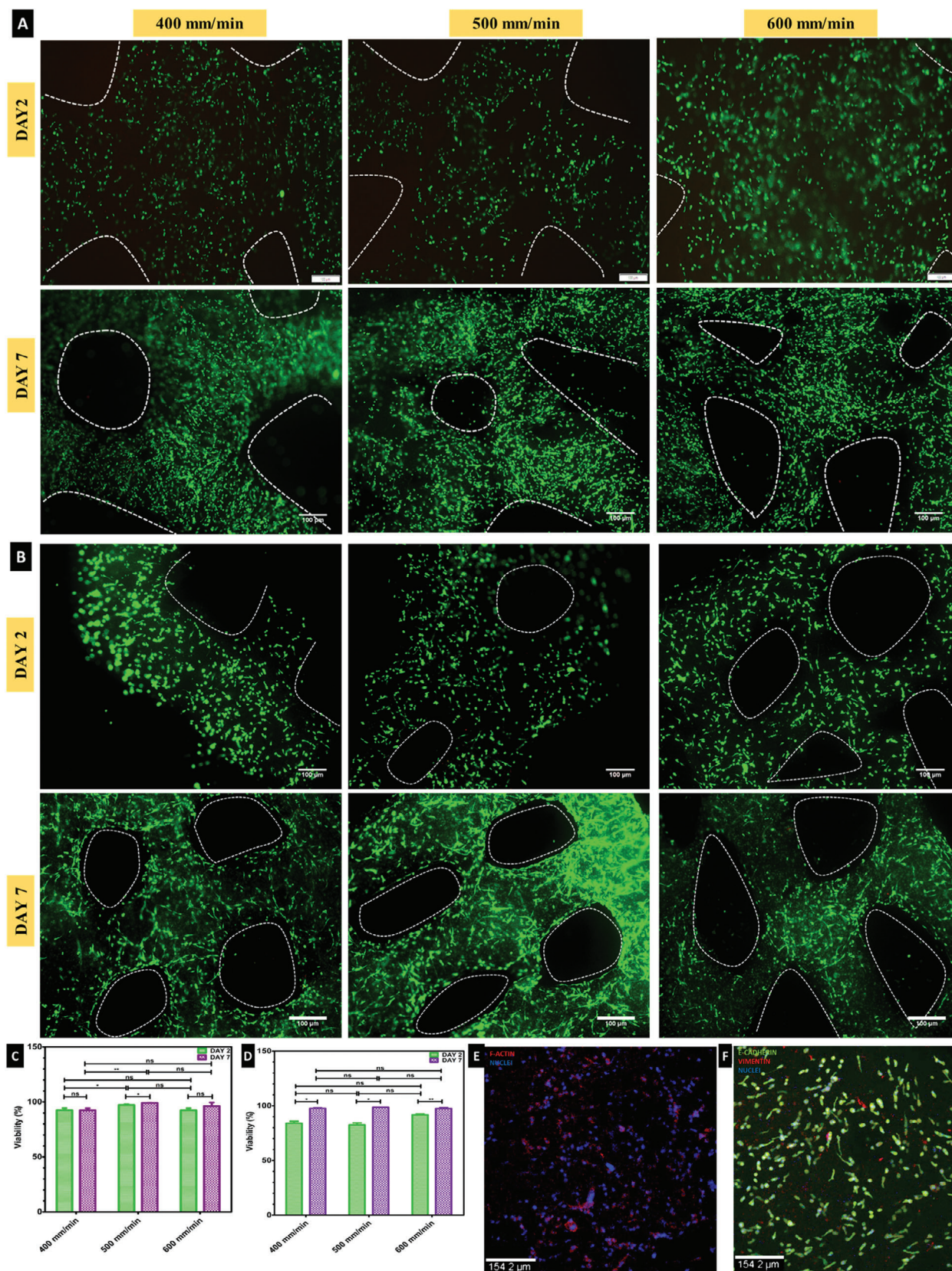


Figure 6. Biological characterization of 3D bioprinted breast cancer model using live/dead assay where live cells are stained green and dead cells are stained red. A) Cell viability of TNBC bioprinted model using MDA-MB-231 laden 3% HH DAT hydrogel based bioink where upper panel denotes day 2

expression of cell surface markers E-Cadherin and Vimentin. It was observed that over 7 days of culture, both round morphology and spindle-shaped MDA-MB-231 cancer cells exhibited both functional markers, as visualized in Figure 6F (separate figures for each marker shown in Figure S5, Supporting Information). These results indicated that the breast cancer cells within bioprinted constructs were present in a conducive microenvironment that could support cancer progression.

2.9. High Throughput 3D Bioprinting of Breast Cancer Model

2.9.1. Fabrication of High Throughput Platform

We wanted to adopt our DAT hydrogel-based 3D bioprinting to develop a scalable, high-throughput in vitro model platform for anti-cancer drug screening. We selected a 96-well plate format and used a 150-micron nozzle laden with 3% DAT hydrogel-based bioink laden with MDA-MB-231 breast cancer cells to fabricate a 3D cancer model using a modified printing approach. Figure 7A showcases the 96-well plate with high throughput printed constructs, while Video S1, Supporting Information showcases the bioprinting process. Figure 7B shows the 3D bioprinted microtumors in all the wells within the 96-well plate. We successfully printed these 3D microtumors within all the wells of a 96-well plate in less than 7 min, a process that, done using a manual pipette, is laborious and time-consuming. Microscopic examination of the printed 3D microtumors revealed the average diameter of constructs to be 1.049 ± 0.233 mm. We compared the diameter of the 3D bioprinted microtumors printed using a 250 μ m nozzle to manually pipetted scaffolds, as shown in Figure 7C, and we also assessed the difference in microtumor diameters across a range of printing pressures, as shown in Figure 7D. Cancer cell-laden 3D bioprinted microtumors at different magnifications can be observed in Figure 7E,F with cells present in a mixed population of round and spread morphology after 24 h of culture. Figure 7G represents the live/dead imaging done on day 2 post-printing to showcase the high cell viability of MDA-MB-231 cells, with the cancer cells showing a spindle-shaped morphology along the matrix showcasing that the cells are in a biomimetic 3D matrix also confirmed through Z stack imaging as shown in Figure 7H where nuclear staining showcased cells populating along the depth of entire 3D printed construct.

2.9.2. Validation of 3D Bioprinted Model by Drug Response Assessment

The bioprinted microtumor constructs were cultured for 3 days in vitro and subjected to 72 h of drug treatment. We chose a widely used chemotherapeutic drug, 5-fluorouracil, and subjected the treated construct to live/dead staining at the endpoint

to detect the viable cell population within the 3D constructs. Figure 7G showcases the number of dead cells stained red gradually increasing as the drug concentration increased compared to the untreated construct. Furthermore, we observed that the number of round and stressed cells increased within printed microtumors exposed to higher concentrations of drugs. The data were quantified in Figure 7H, where a dose-dependent cytotoxic effect of the drug was seen with a significant decrease in the viability of breast cancer cells in drug-treated microtumors as compared to untreated microtumors and a marked chemoresistance to high concentrations of drugs was seen in cancer cells in 3D bioprinted microtumors. Overall, the fabrication of a large number of printed constructs allowed for a better drug screening setup and allowed us to test a wide range of concentrations.

3. Discussion

Successful translational studies require 3D cancer models that can recapitulate the tumor heterogeneity and tumor microenvironment followed by high throughput and high content drug screening.^[46–49] This study presents a method to develop a 3D cancer model with the help of a dECM-based hydrogel derived from a detergent-free decellularization protocol. The proposed process was cost-effective and resourceful for biomaterial sourcing and downstream processing. We were also able to showcase rapid high throughput biofabrication of in vitro 3D drug screening platforms. To recapitulate the breast cancer microenvironment, the biomaterial required for 3D modeling had to be biomimetic in both microarchitecture and microenvironment. Adipose tissue forms a major structural part of the breast, supports mammary ducts and glands,^[50,51] and is also known to play an important role in mammary epithelialization.^[52,53] Therefore, our study utilized decellularized caprine adipose tissue of omental origin to prepare biomimetic hydrogel. Since this study involved ECM preparation from a species that had not been decellularized before, we modified the preexisting adipose tissue detergent-based and detergent-free decellularization method.^[20,34] We performed a combination of decellularization steps (SDS protocol, HH protocol, WH protocol), and did delipidation to ensure maximum cellular proteins and nuclear content removal to prevent unfavorable immunogenic reactions. As per our observations, the detergent-free HH and WH decellularization protocols were more ECM-friendly than the harsher SDS-Triton decellularization process, which has been previously noted elsewhere.^[17,43] The results obtained from SEM imaging showed that the microarchitecture of native tissue was maintained and was in keeping with previously reported ultrastructure.^[18,20,25,33,54]

Biochemical characterizations showed a decrease of hydroxyproline content indicative of total collagen in the matrix as well as sulfated GAGs in DAT, as observed in previous studies.^[25,42,54]

time point, lower panel denotes day 7 time point. B) Cell viability of 3D bioprinted TNBC stromal model using MDA-MB-231 and NIH-3T3 (1:1) laden 3% HH DAT hydrogel based bioink where upper panel denotes day 2 time point, lower panel denotes day 7 time point. Quantification of live/dead assay results comparing cell viability C) in monoculture 3D bioprinted breast cancer at different time points. D) in 3D bioprinted tumor stroma model at different time points. E) Immunofluorescence images showing the expression of functional markers like F-actin, F) E-Cadherin, and Vimentin in encapsulated MDA-MB-231 cells in monoculture constructs on day 7 of culture within 3D bioprinted constructs. (E-Cadherin stained green, Vimentin stained red, nuclear content stained blue). Scale bar size: A,B) 100 μ m and E,F) 154.2 μ m. $n = 3$, $*p \leq 0.05$, $**p \leq 0.01$, and $***p \leq 0.001$.

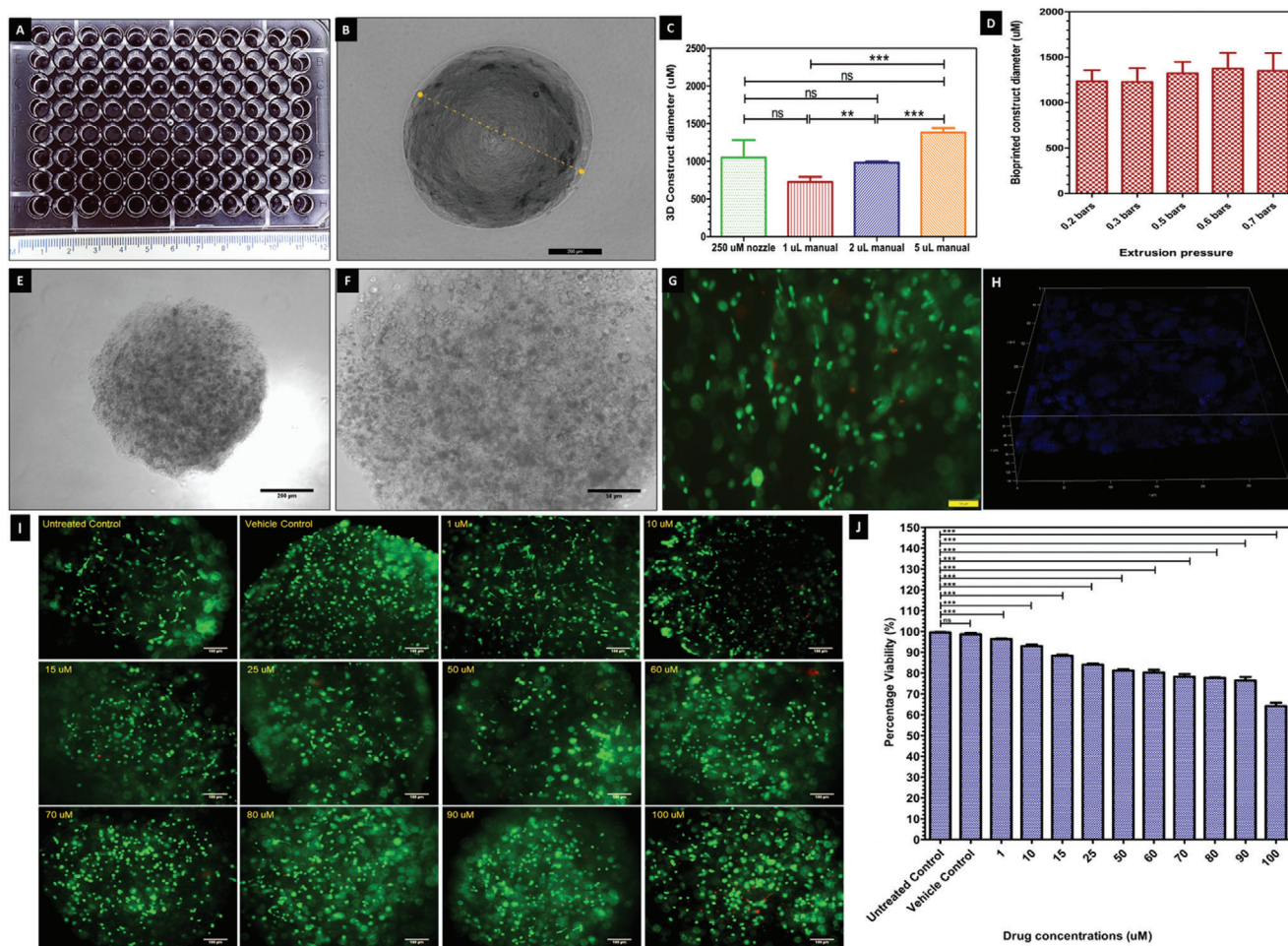


Figure 7. High throughput 3D bioprinting of breast cancer model A) Image of 96-well plate with high throughput 3D bioprinted microtumors. B) Microscopic image of an acellular 3D microtumor. C) Comparison of 3D microtumor diameter attained from 250 μm printing nozzle and manually pipetted scaffolds. D) Optimization of printing pressure parameter for 3D microtumor. E) Microscopic image of bioprinted microtumor post-printing at day 2. F) MDA-MB-231 cells encapsulated within the printed construct. G) Live/dead assay showcasing viable cells post-printing. H) Z stack image of the 3D bioprinted microtumor. I, J) Drug treatment on 3D bioprinted microtumor using 5-fluorouracil. I) Live/dead assay showcasing drug-treated 3D bioprinted microtumors. J) Quantification of viable cells post drug treatment. Scale bar size: B, E) 200 μm, F) 50 μm, G) 100 μm and all images in (J) 100 μm. $n = 3$, $*p \leq 0.05$, $**p \leq 0.01$, and $***p \leq 0.001$.

Due to species differences, our decellularization protocols have a slightly different total hydroxyproline content compared to values reported in human-origin adipose tissue samples in a previous study.^[22] Concerning DNA estimation studies, the initial selection of native adipose tissue samples showed a low DNA amount as the native adipose tissue fraction taken for papain digestion was majorly associated with the white fat component. Therefore, we treated the native adipose tissue samples taken for biochemical assays with IPA for 48 h for uniformity. The resultant native adipose tissue sample showcased a higher DNA quantity during DNA estimation assays, indicating that the samples were digested entirely during papain digestion. Overall, the DNA estimation assay showcased the reduction of DNA content in DAT compared to native adipose tissue, indicating effective decellularization in detergent-based and modified non-detergent-based protocols. These values were well below the previously standardized dsDNA content of 50 ng per mg of decellularized tissue, similar to previous studies.^[17,20,34,43,44] A balance of matrix retention

in the form of collagen content and the least amount of DNA was observed in tissues obtained via the HH protocol, therefore we proceeded with selecting HH and the derivative WH protocol for further experimentations. Before proceeding with cell culture, both Hoechst 33258 staining and ORO staining confirmed that decellularization using HH protocol had effectively removed cellular components as well as lipids and was in keeping with studies before.^[18,19,25,42,54] This was important to ensure no adverse reaction occurred during cell culture experiments. The DAT pre-gel obtained through enzymatic digestion was pH neutralized, and the thermoresponsive hydrogel was crosslinked at 37 °C to form stable 3D scaffolds. Our results indicated that HH DAT and WH DAT hydrogel-based scaffolds were biocompatible and supported cancer cell growth. Since dECM is a natural biomaterial, it is laden with growth factors and native matrix cues that influence its cytocompatibility. DAT hydrogel scaffolds have extensively been used for tissue engineering^[18,23,25,33,34] due to their excellent biocompatibility for a wide range of cells.^[35–40]

The DAT matrix microarchitecture allowed MDA-MB-231 breast cancer cells to grow as isolated cell clusters while a few cells at the edges showed spindle-shaped morphology. Confocal imaging for F-actin, a cytoskeletal protein, showcased that spherical cells were uniformly distributed throughout the 3D scaffold. In keeping with a previously published study, where cancer cells showed a balanced survival dormancy with low-density, small-sized, non-invasive clusters.^[55] This cellular response was seen since HH DAT hydrogel had a normal healthy microenvironment with high crosslinking density and less adhesivity.

We further studied the expression profile for different matrix proteins on encapsulated MDA-MB-231 cells, originating from a non-mammosphere-forming cell line.^[56] Our observation showed low E-Cadherin expression in encapsulated MM321 cells in our hydrogel which led them to remain in smaller, non-invasive clusters. Similar expression profiles have been seen for hydrogels cultured in normoxic conditions for breast cancer cells where high Vimentin levels and low E-Cadherin levels were observed. This is also a survival strategy for metastatic cancer cells to survive in a normoxic environment.^[57,58] Expression for ZO-1 surface receptors, a tight junction protein,^[59,60] was also assessed to be low throughout the culture period. A study showed that when MDA-MB-231 cells were top-seeded and cultured on healthy breast tumor matrix hydrogel for 7 days, they observed low cell clustering with different cell surface receptors showing a differential expression. Therein, Vimentin expression was reported low, and E-Cad and ZO-1 expression increased as cells grew side by side on the 2D top surface.^[40] CD44 receptors are involved in the cell adhesion process, and CD44 expression is also correlated with stemness in breast cancer cells in multiple reports.^[61–64] We observed homogeneous CD44 surface expression with gradual reduction throughout the culture period; however, the proliferation of cells remains the same in vitro, as reported before in a previous study.^[65] We hypothesize that the reduced CD44 expression could be due to multiple factors—first, encapsulation in healthy adipose tissue leading to tumor dormancy; second, the presence of normoxic conditions as shown in some studies,^[57,58] and last due to static culture in our study; as it has been noticed elsewhere that the expression of CD44 isoform increases in response to shear forces.^[66] Overall, our results correlated with cell surface expressions seen for MDA-MB-231 cells in multiple hydrogel systems with different biomaterials.^[67] Furthermore, we also observed that the encapsulated breast cancer cells did not showcase expression for HIF1 α , indicating that cells were not undergoing hypoxic change even after 21 days of culture within the DAT-based scaffold. It is known that in normoxic conditions, HIF1 α will be degraded, while in hypoxic conditions, it will accumulate in the cytoplasm. This shows that the HH DAT scaffolds can maintain normoxic conditions throughout the cell culture period and be used as a biomaterial for long-term longitudinal studies. It is essential to mention that HH DAT hydrogel was derived from natural adipose tissue, the native microenvironment would be more conducive for normal mammary epithelial cell differentiation. Future work in this direction can explore DAT-based hydrogel systems to study the effect of healthy adipose tissue and its interplay with cancer progression. Additionally, introducing dynamic culture to in vitro models might influence the marker expression profile and better mimic in vivo conditions.

We assessed the rheological profile of hydrogel obtained from HH DAT to investigate its potential use for 3D bioprinting purposes. The viscosity-shear rate sweep indicated that HH DAT had a shear thinning property and was in keeping with the viscosity profile of DAT hydrogel observed elsewhere. This shows that hydrogel can be used for extrusion-based 3D bioprinting.^[34,68] The amplitude sweep for HH DAT hydrogel indicated that it had a linear-viscoelastic region, indicating that the gel would be stable during printing and could handle extrusion pressure due to its storage modulus. Similar rheological properties have been noted in studies before for human DAT and porcine fat tissues.^[34,41] Next, we studied the gelation kinetics of the hydrogel to understand the effect of temperature on the hydrogel and the thermoresponsive property for our HH DAT. Our results indicated that thermal crosslinking for HH DAT occurred gradually with a marked increase in complex modulus starting at 23 °C, which provided crucial information regarding the printing parameter of temperature for our hydrogel, which needed to be kept in a lower range to avoid clogging of material during printing. Our results align with a previous study that showed the DAT hydrogel's complex modulus increased with a steady temperature increase.^[34] The DAT hydrogel obtained through our decellularization protocol had a spreading property and could not maintain its shape during the optimization process, but it showcased extrudability at certain printing speeds during test trials. Previous studies have addressed the lack of print fidelity by printing the DAT hydrogel within a mechanically robust predesigned framework or by blending it with additional materials to formulate a bioink that would extrude a continuous filament.^[33,34,41] We used HH DAT hydrogel to encapsulate breast cancer cells and used a different 3D bioprinter to obtain stable 3D bioprinted cancer models. Literature showcasing various 3D bioprinted cancer models fabricated using different biomaterials already exists.^[21,69,70] However, when it comes to using DAT hydrogel for bioprinting 3D cancer models, we did not find preexisting studies using it as a single biomaterial bioink for developing breast cancer models. Chen et al. used porcine DAT hydrogel mixed with GelMA as a viscosity and thermoresponsive modifier to formulate a hybrid bioink laden with MCF7 cells.^[41] Low printability of decellularized mammary tissue-based hydrogel for breast cancer modeling has been shown in another study by Blanco-Fernandez et al., where printing with TDM hydrogel alone was not feasible, and it required two viscosity modifiers GelMA and alginate to improve the printability.^[41] To our knowledge, this study is the first to use DAT-based bioink for extrusion 3D bioprinting without any additional biomaterial. In keeping with the previous studies, our bioprinted cancer model also showcased high cellular viability post-printing, indicating that neither the printing temperature nor the extrusion pressure affected the cell viability.^[41,42] Since adipose tissue forms a significant part of mammary tissue, it is also affected during breast cancer progression, and hence, DAT hydrogel-based bioprinted scaffolds served as an excellent 3D culture system.^[71] In the future, to mimic heterogeneous tumor microenvironment, different cells can be encapsulated and vascularization strategies^[72] can be employed within dECM-based 3D bioprinted cancer models.

Finally, we showed a bioprinting approach where only a minimal amount of bioink was dispensed at a time to fabricate 3D scaffolds in a 96-well plate assay format. One of the most

advantageous applications of 3D bioprinting is that it can rapidly fabricate 3D tissue constructs uniformly in a large number for high throughput screening.^[73] Consequently, this approach reduced the reagent volumes for experimentations like cell culture media and chemotherapeutic drugs. Previously, only a few groups have tried to develop high throughput platforms using 3D bioprinting.^[74–76] The difference in anti-cancer drug response in 2D tissue culture plates (TCP) compared to 3D bioprinted models is well known. While 2D culture is the current gold standard for drug testing, the cells cultured in 2D are phenotypically and genotypically different from those that have undergone bioprinting process and are in 3D microenvironment.^[77,78] For our study, we decided to keep untreated 3D bioprinted microtumors as a control due to their similarity to 3D *in vivo* conditions. We visualized and quantified the cytotoxicity of anti-cancer drugs on 3D microtumors, similar to other studies where only 3D bioprinted scaffolds were characterized for drug response studies.^[79,80] Therefore, our high throughput bioprinted cancer model can be a stepping stone for elevating the quality of drug screening and drug toxicity assay by incorporating the same approach using any type of dECM to develop a personalized cancer models for rapidly screening patient-specific drug responses. Considering the rise of AI to address biomedical problem statements, the future projections lie in using AI-assisted bioprinting of complex, heterogeneous 3D cancer models.^[81]

4. Conclusion

In recent years, there has been an increasing demand for alternative physiologically relevant platforms to reduce the dependency on animal models for drug testing. A critical challenge is producing highly reproducible, less material-consuming, rapidly fabricated screening platforms. 3D bioprinting technology can help fabricate a high throughput platform for anti-cancer drug screening with pre-clinical applications.

In this study, we produced a cheap and robust ECM-based biomaterial from discarded tissue to form DAT-based constructs as 3D scaffolds and 3D bioprinted constructs indicative of their applications for tissue engineering and regenerative medicine. We first effectively decellularized white adipose tissue using a salt-based method compared to the existing detergent-based decellularization protocol and characterized the resultant biomaterials. The results showcased that collagen and sulfated GAGs were highly retained in decellularized tissues obtained via salt-based methods compared to detergent-based protocol, which directly increased the cytocompatibility of the biomaterial. dECM hydrogel was subjected to a rheological study, and its suitability for 3D bioprinting was showcased. Optimization of printing parameters in terms of pressure and printing speed was done to extrude the high-density cell-laden dECM bioink into bilayered lattice constructs. A 3D bioprinted heterogeneous breast cancer model was fabricated laden with triple-negative breast cancer cells and fibroblast cells to replicate the solid breast tumor, followed by thorough characterization of 3D bioprinted constructs to showcase high cellular viability and the expression of functional markers. Furthermore, we modified the printing protocol to generate bioprinted constructs in a high throughput manner for drug screening using the chemotherapeutic drug 5-fluorouracil. In summary, two crucial aspects of decellularization-based tissue engi-

neering were addressed in our study: sourcing dECM-based biomaterials and scaling up 3D bioprinting protocol.

To conclude, this study showcased that DAT hydrogel can be used for 3D bioprinting applications to fabricate biomimetic heterogeneous 3D cancer models with a custom shape and size, laden with multiple cell types which can serve as a pre-clinical tool for translational studies. Our investigations into the feasibility of high throughput bioprinting of 3D microtumors have presented a reliable, user-friendly, rapidly fabricated, and highly reproducible dECM-based platform that can be scaled up easily to study any cancer in a clinical setting and for drug development processes in pharmaceutical industries. Future application of this study involves incorporating patient cancer cells in tissue-specific dECM hydrogel to fabricate a high throughput, patient-specific 3D bioprinted cancer model.

5. Experimental Section

Isolation of Caprine DAT: Caprine adipose tissue of omental origin was obtained from a slaughterhouse. After thoroughly washing the tissue, which often has a distinct network of white fat tissue and the transparent elastic membrane between them, with deionized distilled water, the tissue was minced into small pieces of about 2–3 mm and subjected to different decellularization processes. First, the conventionally used SDS-based decellularization was performed according to the protocol mentioned elsewhere^[34] with few modifications. Briefly, the water-washed tissues were subjected to 1% SDS for 3 days, followed by a wash with 0.1% Triton X-100 for 2 days with continuous stirring. A detergent-free method was also employed for adipose tissue decellularization, herewith referred to as “Hypo-Hyper” based decellularization, which was performed by washing tissues with the salt solutions alternating between hypertonic 1 M NaCl solution and hypotonic 0.1 M NaCl solution with distilled water wash in between for 5 min. A similar modification to the HH protocol was made by alternating between 1 M NaCl and distilled water for 5 days, with distilled water wash for 5 min in between each change of decellularizing solution, herewith referred to as “Distilled Water-Hyper” based decellularization. After the decellularization steps, tissues were again washed with deionized distilled water to remove any traces of residual detergents. Decellularized tissues obtained via all protocols were then subjected to 48 h of 100% IPA treatment for delipidation, along with a batch of native tissue of the same original weight as decellularized tissue for further downstream experiments. This was followed again by washing with deionized distilled water. To prevent contamination of decellularized tissues from fungal spores, they were subjected to an overnight 3% H₂O₂ wash followed by 3 days of distilled water wash to remove any ions left within the matrix. Finally, the decellularized tissues were subjected to lyophilization for 48 h and stored at –20 °C until further use.

SEM Assessment: Lyophilized fibers of decellularized tissues obtained from the SDS-based protocol, HH protocol, and WH protocol were subjected to surface coating. Samples were sputter-coated with gold using a thin film deposition system (Kurt J. Lesker CMS-18 HV) for 30 s with a 25 mA electric. Representative images were taken from one of the seven batches of prepared DAT, with four images from different sections of each test sample. SEM images were procured on a field emission scanning electron microscope (JOEL JSM-7800F Japan) at two different magnifications in vacuum conditions while operating under an accelerated voltage of 15 kV.

Biochemical Characterization of Decellularized Tissue: The decellularized and native adipose tissue treated with IPA for 48 h was used to estimate DNA, sulfated GAG, and hydroxyproline collagen contents. IPA wash was introduced for native adipose tissue, as the presence of lipid hampers papain digestion. This step led to complete papain digestion required for biochemical assays. For the biochemical estimations, decellularized and native tissues were digested in 1 mL papain solution for 16 h till the complete tissue was digested without any residues at 60 °C following the

protocol described elsewhere.^[34] Briefly, the papain solution contained 0.125 mg mL⁻¹ papain in 0.1 M sodium phosphate with 5 mM EDTA and 5 mM cysteine-HCl at pH 6.5.

DNA Estimation: To evaluate the extent of decellularization for each decellularization batch, DNA was measured with the PicoGreen quantitative assay kit (Invitrogen, UK). The fluorescence was measured using a spectrophotometer (PerkinElmer, UK). Quantitative measurement for DNA was calculated from the equation obtained from a standard curve generated using calf thymus DNA, with data presented as mean $\pm$ SD.

Collagen and GAG Estimation: GAG content was estimated using a 1,9-dimethyl methylene blue solution. A plate reader was used to measure the absorbance of the samples, which were read at 492 nm. Quantitative measurement was calculated from the standard curve obtained from the chondroitin sulfate standard, and data presented as mean $\pm$ SD. ECM component collagen that was retained in the decellularized tissue was estimated using a hydroxyproline assay. Collagen quantification was done using dimethyl amino benzaldehyde (Sigma Aldrich, USA) and chloramine-T (Sigma Aldrich, USA) as per the manufacturer's protocol, and absorbance was noted at 570 nm, data presented as mean $\pm$ SD.

Normalization of Tissue Weights: Usually, for biochemical assays, equal weights of native and decellularized tissues are taken for papain digestion, and later on, the estimated value of decellularized tissue is normalized to the estimated value of native tissue. However, since the decellularized tissue was devoid of cellular content, the percentage of ECM components will increase per unit volume of the decellularized tissue, which can give rise to a false estimation of its residual ECM components. The native tissue contains cells and other water-soluble materials that will be lost during decellularization. Hence, the % weight loss method was calculated and that value was used as a factor for further calculation. The yield percentage was calculated using the following formula and has been used elsewhere.^[82]

$$\text{Yield \%} = 100 - ((\text{weight of dry lyophilized native tissue} - \text{weight of dry lyophilized decellularized tissue}) / \text{weight of dry lyophilized native tissue}) \times 100$$

The yield % value was used to calculate retained ECM in decellularized tissues obtained in this study with respect to native tissue to calculate the results of the assays for collagen and GAG estimation.

Histological and Immunostaining Analysis: Native adipose tissue and wet HH decellularized tissue (pre-lyophilization) were fixed in 4% formalin, embedded in OCT compound (ThermoFisher Scientific, US), and then taken for cryosectioning. Sections of 10-micron thickness were generated for Hoechst 33258 staining (Invitrogen, UK) to analyze nuclear content in native and decellularized tissue sections. Oil Red O staining was performed on wet native and decellularized tissue. 5 mg mL⁻¹ Oil Red O stain (Sigma Aldrich, US) was first prepared using 100% isopropanol, which was further diluted with distilled water in a 6:4 ratio. The stain was centrifuged, and the supernatant was used to stain tissue samples for 5 min, followed by multiple washes with distilled water. The sections were then imaged under a microscope.

Hydrogel Preparation: For this study, the focus was on 3% w/v DAT hydrogel, which was prepared by weighing a sufficient amount of lyophilized tissue and solubilizing it in a solution containing 0.5 M acetic acid with 10% (w/w) pepsin with respect to the weight of the tissue. After solubilization for 24 h for SDS-based decellularized tissue and 48 h for NaCl-based decellularized tissues, the pH of the pregel was adjusted to nearly 7.4 by adding cold 10 M of NaOH. During the process of pH adjustment, the pregel was always kept at a cold temperature to avoid gelation. The pH-adjusted hydrogel was stored in a refrigerator at 4 °C, and each batch was used for experimentation within a week of preparation. Thermal crosslinking at 37 °C was used to form stable 3D hydrogel scaffolds from the hydrogel. The crosslinking time was optimized for DAT hydrogels by checking their stability in PBS and was kept uniformly for 45 min of thermal crosslinking for all the experiments in the study.

Rheological Assessment: pH-adjusted 3% DAT hydrogel was subjected to rheological studies using the rheometer Kinexus Pro+ (Malvern Panalyticals). A temperature sweep was performed with a ramp rate of 0.5 °C min⁻¹ at a gap of 0.3 mm with shear strain fixed at 0.01 and a frequency of 1 Hz to understand the gelation kinetics of DAT hydrogel. A time sweep experiment was performed to understand the dynamic moduli

of the hydrogel at a crosslinking temperature of 37 °C, with the measurements taken for an incubation time of 60 min. An amplitude sweep was conducted on uncrosslinked DAT hydrogel at a shear strain range of 0.01–100% at a temperature of 20 °C with a gap of 0.3 mm and frequency fixed at 1 Hz. A shear sweep from 0.01 to 1000/s was performed on the DAT hydrogel at a temperature of 20 °C with a gap of 0.3 mm and frequency fixed at 1 Hz to measure the DAT hydrogel viscosity.

Cell Culture: MDA-MB-231 breast cancer cells were cultured in Minimal eagle medium (Lonza, Europe), NIH-3T3 embryonic mouse fibroblast cells were cultured in DMEM supplemented (Himedia, India), 4T1 breast cancer cells were cultured in RPMI (Himedia, India) and primary limbal cells were culture in α MEM (Himedia, India) with 10% fetal bovine serum (Biowest, France) in 37 °C, 5% CO₂ humidified environment. Confluent cells were removed from the TCP with 0.25% trypsin solution, centrifuged at 1800 rpm for 5 min, suspended in 1 mL fresh 1 X MEM media, and counted. For the encapsulation study, the pH-adjusted DAT pregel was mixed with 10% of 10 $\times$ concentrated culture media (Sigma Aldrich, USA), and suspended cells at a density of 1 $\times$ 10⁷ cells mL⁻¹ were added to the DAT pregel and mixed thoroughly. The cell-encapsulated DAT hydrogel was kept at 37 °C for thermal crosslinking in a 5% CO₂ humidified environment for 45 min to allow the hydrogel to undergo gelation.

Cell Viability and Metabolic Activity Evaluation: Calcein-AM (Invitrogen, UK) and propidium iodide probe (Sigma Aldrich, Germany) were used to perform live/dead assay to identify live cells (stained green) and dead cells (stained red), respectively, based on the cell membrane integrity. Live/dead was performed on the sample at different time points on days 1 and 7. Independent biological replicates in distinct experiments were stained, and a representative image from one set has been shown. At time points day 1, 4, and 7 of cell culture, an MTT assay was conducted for metabolic activity analysis on cell-encapsulated 3D scaffolds. MTT reagent (Sigma Aldrich) was added to the sample and was incubated for 3 h followed by incubation in DMSO solution for 30 min to solubilize formazan crystals formed. The soft hydrogel scaffolds were mechanically crushed to ensure all the formazan crystals came out and solubilized in the DMSO solution, and the absorbance was measured using a plate reader at a wavelength of 570 nm.

Immunofluorescence Assessment: Breast cancer cell-laden DAT constructs were fixed after days 1 and 7 of culture and then stained with F-actin (Invitrogen, UK) and Hoechst 33258 (Invitrogen, UK) for confocal imaging to assess the cellular morphology. To assess the cellular functionality of encapsulated MDA-MB-231 breast cancer cells, the cell-laden DAT hydrogel constructs were fixed in 10% formaldehyde at 7, 14, and 21 days of culture and imaged. Fixed samples were stained with the markers E-Cadherin (4A2) Mouse mAb #14472 (Cell Signaling Technology, USA), Vimentin (D21H3) XP Rabbit mAb (Cell Signaling Technology, USA), ZO-1 Monoclonal Antibody (ZO1-1A12) (Thermo Fisher Scientific, USA), and CD44 Polyclonal Antibody (Invitrogen, UK), as well as Hoechst 33258 (Invitrogen, UK), was used immunofluorescence experiments. Primary antibodies were used in a concentration of 1:500, while secondary antibodies, anti-rabbit Alexa Fluor 594 (Cell Signaling Technology, USA) and anti-mouse Alexa Fluor Plus 647 (Invitrogen, UK), were used in a concentration of 1:200.

Extrudability of DAT Hydrogel for 3D Printing: A commercial 3D printing equipment, BIO V1 from Regemat 3D (a screw-driven extrusion-based 3D printing), was used for the 3D printing process. The .stl file was made and modified on the company's trademark printer software, Regemat 3D Designer, and the lattice model (15 $\times$ 15 $\times$ 1 mm) was uploaded for printing. A syringe barrel fitted with a Nordson General Purpose 25 G tapered nozzle (needle with an internal diameter of 250 μ m) and a Nordson General Purpose 27 G tapered nozzle (needle with an internal diameter of 410 μ m) was filled with filtered 3% HH DAT hydrogel and loaded into the cold extruder tool head at a temperature of 15 °C. Extrusion-based 3D printing was done layer-by-layer while testing different nozzle diameters and flow speeds. Printing was done at room temperature (RT) of 25 °C and optimized based on its rheological data. The printed construct was then analyzed for strand fidelity by microscopic analysis.

Bioink Preparation, Optimization of Printing Parameters, and 3D Bio-printing: Once the 3% DAT hydrogel extrudability was considered fit for

extrusion-based printing, bioink extrudability and bioprinting were optimized using a proprietary printer from Next Big Innovation Labs. All the components, like the nozzle, syringe barrel, and plunger, were sterilized with 70% ethanol and kept in a laminar hood under UV light for 15 min, followed by washing with autoclaved distilled water before filling it with DAT hydrogel-based bioink for bioprinting. Bioink preparation was the same as hydrogel preparation and done over ice to prevent crosslinking, with cell suspension added at a seeding density of 10 million cells mL⁻¹ hydrogel. Air bubbles were removed from the syringe barrel using centrifugation.

The first geometry tested was a 15 × 15 × 1 mm³ lattice structure. The syringe barrel fitted with a Nordson General Purpose Stainless Steel Tip 25 G nozzle (needle with an internal diameter of 250 μm) was filled with cell-laden 3% HH DAT hydrogel and loaded into the cold extruder tool head with the printing temperature set to 15 °C. MDA-MB-231 breast cancer cells and NIH-3T3 fibroblasts were used for bioprinting cancer models—bioinks compromising of just cancer cell-laden DAT hydrogel to form monoculture model or bioinks with both cancer cells and fibroblasts laden DAT hydrogel in a ratio of 2:1 to form coculture model. Both the cellular inks were separately tested and assessed for process optimization and cytocompatibility. Printing pressure was set at 0.3 bars while different printing speeds of 400, 500, and 600 mm min⁻¹ were tested.

A modified printing approach was employed within a 96-well format for high-throughput bioprinting. A barrel fitted with a Nordson General Purpose Stainless Steel Tip 30 G nozzle (needle with an internal diameter of 150 μm) was filled with MDA-MB-231 breast cancer cells laden 3% HH DAT hydrogel (seeding density 10 million cells mL⁻¹ hydrogel) and loaded into the cold extruder tool head. The printing was conducted at a temperature of 15 °C at a speed of 800 mm min⁻¹, and the printing pressure was set at 0.3 bars. G-Code has been provided in supporting information. Both acellular and cellular inks were tested for process optimization and assessed for cytocompatibility within the 3D microtumors.

All printed constructs were incubated for 30 min at 37 °C for thermal crosslinking, and cell culture media was added subsequently. Printed constructs were characterized using a bright field microscope to check for strand fidelity, and cell viability for bioprinted constructs was assessed using the live/dead assay method described in Section 2.8.

Assessment of 5-Fluorouracil on Bioprinted 3D Microtumors: Validation of 3D bioprinted microtumors was done by exposing the breast cancer microtumors to drug treatment using the anti-cancer drug 5-fluorouracil. A stock solution was prepared in DMSO, and all the drug concentrations were made by diluting the stock solution in MEM. The 3D bioprinted microtumors were exposed to 5-FU concentrations at 1, 10, 15, 25, 50–100 μM. The final concentration of the DMSO was less than 1.5% of the highest 5-FU drug concentration. The duration of drug treatment was 72 h, and the drug cytotoxicity was evaluated within the 3D microtumors using live/dead Assay using Calcein-AM to stain live cells green and PI to stain dead cells, as described in Section 2.8.

Statistical Analysis: Unless mentioned, all the experiments were conducted in triplicates ($n = 3$), and data are represented as mean ± standard deviation. All statistical analysis was performed on GraphPad Prism software, and significance was checked using the student t-test and ANOVA. * $p \leq 0.05$, ** $p \leq 0.01$, and *** $p \leq 0.001$ were considered statistically significant, and ns indicates statistically non-significant.

Ethical Statement: No human subjects, human data, or tissue were used for this study, and no animal testing was conducted.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3D bioprinting, adipose tissue, breast cancer, decellularized extracellular matrix hydrogel, high throughput

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- [1] S. V. Murphy, A. Atala, *Nat. Biotechnol.* **2014**, *32*, 773.
- [2] M. A. Heinrich, W. Liu, A. Jimenez, J. Yang, A. Akpek, X. Liu, Q. Pi, X. Mu, N. Hu, R. M. Schifferers, J. Prakash, J. Xie, Y. S. Zhang, *Small* **2019**, *15*, 1805510.
- [3] N. Ashammakhi, S. Ahadian, C. Xu, H. Montazerian, H. Ko, R. Nasiri, N. Barros, A. Khademhosseini, *Mater. Today Bio* **2019**, *1*, 100008.
- [4] C. Walker, E. Mojares, A. Del Río Hernández, *Int. J. Mol. Sci.* **2018**, *19*, 3028.
- [5] A. Gilpin, Y. Yang, *Biomed. Res. Int.* **2017**, *1*, 9831534.
- [6] H. Lee, W. Han, H. Kim, D. H. Ha, J. Jang, B. S. Kim, D. W. Cho, *Biomacromolecules* **2017**, *18*, 1229.
- [7] E. Garreta, R. Oria, C. Tarantino, M. Pla-Roca, P. Prado, F. Fernández-Avilés, J. M. Campistol, J. Samitier, N. Montserrat, *Mater. Today* **2017**, *20*, 166.
- [8] M. M. De Santis, H. N. Alsafadi, S. Tas, D. A. Bölükbas, S. Prithiviraj, I. A. Da Silva, M. Mittendorfer, C. Ota, J. Stegmayr, F. Daoud, M. Königshoff, K. Sward, J. A. Wood, M. Tassieri, P. E. Bourguine, S. Lindstedt, S. Mohlin, D. E. Wagner, *Adv. Mater.* **2021**, *33*, 2005476.
- [9] S. S. Choe, J. Y. Huh, I. J. Hwang, J. I. Kim, J. B. Kim, *Front. Endocrinol.* **2016**, *7*, 30.
- [10] E. C. M. Mariman, P. Wang, *Cell. Mol. Life Sci.* **2010**, *67*, 1277.
- [11] M. Spencer, R. Unal, B. Zhu, N. Rasouli, R. E. McGehee, C. A. Peterson, P. A. Kern, *J. Clin. Endocrinol. Metab.* **2011**, *96*, 1990.
- [12] M. Soták, M. R. Rajan, M. Clark, C. Börserud, V. Wallenius, C. E. Hagberg, E. Börgeson, *Int. J. Mol. Sci.* **2022**, *23*, 520.
- [13] L. H. Corrêa, G. S. Heyn, K. G. Magalhaes, *Cells* **2019**, *8*, 662.
- [14] N. Alkhouli, J. Mansfield, E. Green, J. Bell, B. Knight, N. Liversedge, J. C. Tham, R. Welbourn, A. C. Shore, K. Kos, C. P. Winlove, *Am. J. Physiol. – Endocrinol. Metab.* **2013**, *305*, 1427.
- [15] S. S. Sharath, J. Ramu, S. V. Nair, S. Iyer, U. Mony, J. Rangasamy, *Tissue Eng. Regen. Med.* **2020**, *17*, 123.
- [16] C. M. Mahoney, C. Imbarlina, C. C. Yates, K. G. Marra, *Front. Pharmacol.* **2018**, *9*, 507.
- [17] D. A. Banyard, V. Borad, E. Amezcua, G. A. Wirth, G. R. D. Evans, A. D. Widgerow, *Aesthetic Surg. J.* **2016**, *36*, 349.
- [18] L. Wang, J. A. Johnson, Q. Zhang, E. K. Beahm, *Acta Biomater.* **2013**, *9*, 8921.
- [19] B. N. Brown, J. M. Freund, L. Han, J. P. Rubin, J. E. Reing, E. M. Jeffries, M. T. Wolf, S. Tottey, C. A. Barnes, B. D. Ratner, S. F. Badylak, *Tissue Eng., Part C* **2011**, *17*, 411.

- [20] L. E. Flynn, *Biomaterials* **2010**, 31, 4715.
- [21] D. Choudhury, H. W. Tun, T. Wang, M. W. Naing, *Trends Biotechnol.* **2018**, 36, 787.
- [22] O. A. Mohiuddin, B. Campbell, J. N. Poche, C. Thomas-Porch, D. A. Hayes, B. A. Bunnell, J. M. Gimble, *Adv. Exp. Med. Biol.* **2020**, 1212, 57.
- [23] Y. C. Choi, J. S. Choi, B. S. Kim, J. D. Kim, H. I. Yoon, Y. W. Cho, *Tissue Eng., Part C* **2012**, 18, 866.
- [24] J. S. Choi, B. S. Kim, J. Y. Kim, J. D. Kim, Y. C. Choi, H. J. Yang, K. Park, H. Y. Lee, Y. W. Cho, *J. Biomed. Mater. Res., Part A* **2011**, 97, 292.
- [25] M. Song, Y. Liu, L. Hui, *Mol. Med. Rep.* **2018**, 17, 138.
- [26] B. H. Clough, M. R. McCarley, U. Krause, S. Zeitouni, J. J. Froese, E. P. McNeill, C. D. Chaput, H. W. Sampson, C. A. Gregory, *J. Bone Miner. Res.* **2015**, 30, 83.
- [27] V. Guneta, Z. Zhou, N. S. Tan, S. Sugii, M. T. C. Wong, C. Choong, *Biomater. Sci.* **2018**, 6, 168.
- [28] S. M. H. Haddad, E. Omid, L. E. Flynn, A. Samani, *J. Mech. Behav. Biomed. Mater.* **2016**, 57, 235.
- [29] I. Elgali, O. Omar, C. Dahlin, P. Thomsen, *Eur. J. Oral Sci.* **2017**, 125, 315.
- [30] T. T. Y. Han, S. Toutounji, B. G. Amsden, L. E. Flynn, *Biomaterials* **2015**, 72, 125.
- [31] V. E. Getova, J. A. van Dongen, L. A. Brouwer, M. C. Harmsen, *Artif. Cells, Nanomed., Biotechnol.* **2019**, 47, 1693.
- [32] C. J. Poon, M. V. Maria, S. Sinha, J. A. Palmer, A. A. Woods, W. A. Morrison, K. M. Abberton, *Acta Biomater.* **2013**, 9, 5609.
- [33] F. Pati, D. H. Ha, J. Jang, H. H. Han, J. W. Rhie, D. W. Cho, *Biomaterials* **2015**, 62, 164.
- [34] F. Pati, J. Jang, D. H. Ha, S. Won Kim, J. W. Rhie, J. H. Shim, D. H. Kim, D. W. Cho, *Nat. Commun.* **2014**, 5, 3935.
- [35] R. Castelló-Cros, D. R. Khan, J. Simons, M. Valianou, E. Cukierman, *BMC Cancer* **2009**, 9, 1.
- [36] T. Hoshiba, M. Tanaka, *Biochem. Biophys. Res. Commun.* **2013**, 439, 291.
- [37] L. W. Dunne, Z. Huang, W. Meng, X. Fan, N. Zhang, Q. Zhang, Z. An, *Biomaterials* **2014**, 35, 4940.
- [38] Q. Jin, G. Liu, S. Li, H. Yuan, Z. Yun, W. Zhang, S. Zhang, Y. Dai, Y. Ma, *J. Cell. Physiol.* **2019**, 234, 3425.
- [39] G. Liu, B. Wang, S. Li, Q. Jin, Y. Dai, *J. Cell. Physiol.* **2019**, 234, 9447.
- [40] K. F. Ruud, W. C. Hiscox, I. Yu, R. K. Chen, W. Li, *Breast Cancer Res.* **2020**, 22, 82.
- [41] Y. Chen, L. Xu, W. Li, W. Chen, Q. He, X. Zhang, J. Tang, Y. Wang, B. Liu, J. Liu, *Biofabrication* **2022**, 14, 025002.
- [42] B. Blanco-Fernandez, S. Rey-Vinolas, G. Bağcı, G. Rubi-Sans, J. Otero, D. Navajas, S. Perez-Amodio, E. Engel, *ACS Appl. Mater. Interfaces* **2022**, 14, 29467.
- [43] J. E. Reing, B. N. Brown, K. A. Daly, J. M. Freund, T. W. Gilbert, S. X. Hsiong, A. Huber, K. E. Kullas, S. Tottey, M. T. Wolf, S. F. Badylak, *Biomaterials* **2010**, 31, 8626.
- [44] P. M. Crapo, T. W. Gilbert, S. F. Badylak, *Biomaterials* **2011**, 32, 3233.
- [45] J. Emmermacher, D. Spura, J. Cziommer, D. Kilian, T. Wollborn, U. Fritsching, J. Steingroewer, T. Walther, M. Gelinsky, A. Lode, *Biofabrication* **2020**, 12, 025022.
- [46] C. H. Wong, K. W. Siah, A. W. Lo, *Biostatistics* **2019**, 20, 273.
- [47] E. Fröhlich, *Int. J. Mole. Sci.* **2023**, 24, 7116.
- [48] Y. Kang, P. Datta, S. Shanmughapriya, I. T. Ozbolat, *ACS Appl. Bio Mater.* **2020**, 3, 5552.
- [49] Y. Shrike Zhang, M. Duchamp, R. Oklu, L. W. Ellisen, R. Langer, A. Khademhosseini, *ACS Biomater. Sci. Eng.* **2016**, 2, 1710.
- [50] D. T. Ramsay, J. C. Kent, R. A. Hartmann, P. E. Hartmann, *J. Anat.* **2005**, 206, 525.
- [51] D. Devadas, T. A. Moore, N. Walji, E. W. K. Young, *Biomicrofluidics* **2019**, 13, 064122.
- [52] X. Hu, S. C. Juneja, N. J. Maihle, M. P. Cleary, *J. Natl. Cancer Inst.* **2002**, 94, 1704.
- [53] B. A. Nerger, C. M. Nelson, *Biomaterials* **2019**, 198, 135.
- [54] G. Zhang, H. Ci, C. Ma, Z. Li, W. Jiang, L. Chen, Z. Wang, M. Zhou, J. Sun, *Acta Biomater.* **2022**, 148, 90.
- [55] S. Pradhan, J. H. Slater, *Biomaterials* **2019**, 215, 119177.
- [56] J. M. Iglesias, I. Beloqui, F. Garcia-garcia, O. Leis, A. Vazquez-martin, A. Eguirra, S. Cufi, A. Pavon, J. A. Menendez, J. Dopazo, A. G. Martin, Mammosphere Formation in Breast Carcinoma Cell Lines Depends upon Expression of E-cadherin, **2013**, 8, e77281.
- [57] Y. Wang, S. Mirza, S. Wu, J. Zeng, W. Shi, H. Band, V. Band, B. Duan, *Oncotarget* **2018**, 9, 32191.
- [58] M. Pal, H. Chen, B. H. Lee, J. Y. H. Lee, Y. S. Yip, N. S. Tan, L. P. Tan, *Sci. Rep.* **2019**, 9, 8997.
- [59] E. A. Rakha, J. S. Reis-filho, F. Baehner, D. J. Dabbs, T. Decker, V. Eusebi, S. B. Fox, S. Ichihara, J. Jacquemier, S. R. Lakhani, J. Palacios, A. L. Richardson, S. J. Schnitt, F. C. Schmitt, P.-H. Tan, G. M. Tse, S. Badve, I. O. Ellis, *Breast Cancer Res.* **2010**, 12, 207.
- [60] K. B. Hoover, S. Y. Liao, P. J. Bryant, *Am. J. Pathol.* **1998**, 153, 1767.
- [61] S. Goodison, V. Urquidí, D. Tarin, *J. Clin. Pathol.* **1999**, 52, 189.
- [62] J. R. C. Hinke, A. B. Mulhaupt, B. Leitinger, D. Gullberg, *Adv. Drug Delivery Rev.* **2016**, 97, 28.
- [63] A. Ouhtit, S. Madani, I. Gupta, S. Shanmuganathan, M. E. Abdraboh, H. Al-Riyami, Y. M. Al-Farsi, M. H. G. Raj, *J. Cancer* **2013**, 4, 566.
- [64] L. T. Senbanjo, M. A. Chellaiah, *Front. Cell Dev. Biol.* **2017**, 5.
- [65] S. McFarlane, J. A. Coulter, P. Tibbitts, A. O'Grady, C. McFarlane, N. Montgomery, A. Hill, H. O. McCarthy, L. S. Young, E. W. Kay, C. M. Isacke, D. J. J. Waugh, *Oncotarget* **2015**, 6, 11465.
- [66] V. S. Shirure, T. Liu, L. F. Delgadillo, C. M. Cuckler, D. F. J. Tees, F. Benencia, D. J. Goetz, M. M. Burdick, *Am. J. Physiol.: Cell Physiol.* **2015**, 308, C68.
- [67] P. S. Thakuri, C. Liu, G. D. Luker, H. Tavana, *Adv. Healthcare Mater.* **2018**, 7, 1.
- [68] M. T. Wolf, K. A. Daly, E. P. Brennan-Pierce, S. A. Johnson, C. A. Carruthers, A. D'Amore, S. P. Nagarkar, S. S. Velankar, S. F. Badylak, *Biomaterials* **2012**, 33, 7028.
- [69] M. Tamayo-Angorrilla, J. López de Andrés, G. Jiménez, J. A. Marchal, *Transl. Res.* **2022**, 247, 117.
- [70] J. Bae, S. Han, S. Park, *BioChip J.* **2020**, 14, 137.
- [71] C. Kothari, C. Diorio, F. Durocher, *Int. J. Mol. Sci.* **2020**, 21, 5760.
- [72] S. Chae, D. H. Ha, H. Lee, *Int. J. Bioprint.* **2023**, 748.
- [73] A. Mazzocchi, S. Soker, A. Skardal, *Appl. Phys. Rev.* **2019**, 6, 011302.
- [74] K. Ling, G. Huang, J. Liu, X. Zhang, Y. Ma, T. Lu, F. Xu, *Engineering* **2015**, 1, 269.
- [75] T. J. Puls, X. Tan, M. Husain, C. F. Whittington, M. L. Fishel, S. L. Voytik-Harbin, *Sci. Rep.* **2018**, 8.
- [76] E. Maloney, C. Clark, H. Sivakumar, K. Yoo, J. Aleman, S. A. P. Rajan, S. Forsythe, A. Mazzocchi, A. W. Laxton, S. B. Tatter, R. E. Strowd, K. I. Votanopoulos, A. Skardal, *Micromachines* **2020**, 11, 208.
- [77] L. Sun, H. Yang, Y. Wang, X. Zhang, B. Jin, F. Xie, Y. Jin, Y. Pang, H. Zhao, X. Lu, X. Sang, H. Zhang, F. Lin, W. Sun, P. Huang, Y. Mao, *Front. Oncol.* **2020**, 10, 878.
- [78] A. Campbell, J. E. Mohl, D. A. Gutierrez, A. Varela-Ramirez, T. Boland, *Front. Bioeng. Biotechnol.* **2020**, 8, 82.
- [79] Y. Wang, W. Shi, M. Kuss, S. Mirza, D. Qi, A. Krasnoslobodtsev, J. Zeng, H. Band, V. Band, B. Duan, *ACS Biomater. Sci. Eng.* **2018**, 4, 4401.
- [80] E. M. Langer, B. L. Allen-Petersen, S. M. King, N. D. Kendsersky, M. A. Turnidge, G. M. Kuziel, R. Riggers, R. Samatham, T. S. Amery, S. L. Jacques, B. C. Sheppard, J. E. Korkola, J. L. Muschler, G. Thibault, Y. H. Chang, J. W. Gray, S. C. Presnell, D. G. Nguyen, R. C. Sears, *Cell Rep.* **2019**, 26, 608.
- [81] H. Lee, *Cyborg Bionic Syst.* **2023**, 4, 0018.
- [82] A. K. Bera, Y. Sriya, F. Pati, *Macromol. Biosci.* **2022**, 22, 2200109.